ROYAL CANADIAN NAVY
QUALIFICATION STANDARD
AND PLAN (QSP)

MARINE TECHNICIAN (MAR TECH)
(MOSID 00379)

RQ-PO1 (ALUV)

MITE Course IDs:
MAR TECH RQ-PO1 – 121848
DRMIS Course Mod 3 (Maintenance Manager) – 119098

RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commanding Officer of the Naval Training Development Centre (NTDC). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J "Course History".

Revision	Approval Date	Authority
1.0 (QSPWB March 2018)		NTDC(P) CO
		CNPTG
1.1	14/06/2024	CNTG

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FOREWORD

1. This training document is developed under the authority of the Commander Naval Personnel and Training Group (CNPTG) and in accordance with the following references:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-002/PP-002 Non-Commissioned Member General Specification dated 9 Feb 11;

   c. A-PD-055-002/PP-002 Marine Technician Job Based Occupational Specification dated 4 Apr 17;

   d. NAVORD 4500-0 - Structured Delegation; and

   e. NPTORD 4500-22 – Qualification Standard and Plan Production and Management.

2. This QSP consists of specific Performance Objectives (POs), Enabling Objectives (EOs), and Lesson Specifications in Chapter 3. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA (NTDC(P) MS div), who will forward to NPTG for approval, as appropriate.

5. The designated TEs for this training are indicated below:

TE	Phase / Module	MITE Course ID	NQual
NFS(P)	All	121848	ALUV
NFS(A)			

CHAPTER 1 TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL TRAINING GROUP (CNTG)

MARPAC RDIMS #793521
Chap, 1, approved 14 06 2024
Chap 2, 3 approved 14 06 2024

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the MAR TECH manager. This QSP is designed to assist training establishments in developing, conducting, evaluating and validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22- Qualification Standard and Plan, Annex B.

Allocation of ATT	Minutes
Sum of all EOs/POs	8625
Enabling Checks (ECs)	2760
Performance Checks (PCs)	485
Actual Training Time (ATT) Sub-total	**11870**
Allocation of BAT, TAT, TTT, PT (Course < 31 days)	
Base Administrative Time (BAT)	540
Training Administrative Time (TAT)	480
Training Travel Time (TTT)	60
Physical Fitness (PT)	900
Sub-total	**1980**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT)	**13850**
Total mins/400 = # days (rounded up)	
Training Days	**35**

RESOURCES

3. The following are key personnel required to conduct the training.

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO / EO
Sr Instructor	CPO2	MAR TECH RQ-PO1	1	PO 002-004, 006
Instructor	PO1	MAR TECH RQ-PO1 (Mechanical)	1	PO 002-004,006
Instructor	PO1	MAR TECH RQ-PO1 (Electrical)	1	PO 002 – 004,006
DRMIS Instructor	Civ / PO1/ CPO2	DRMIS Maintenance Supervisor Course Manager Course	2	PO 001

Dock Master/ Officer	Civ	FMFCB / FMFCS Docking Officer	1	EO 003.01
Proj Mgmt Instructor	Civ / PO1 or above	Project Management Professional (PMP) Certification	1	PO 005
Technical SME	PO1 or above	MAR TECH RQ-PO1 or NTO HOD (AISE)	1	PO 005

4. The following table details the critical training resources without which the training cannot be conducted. It should be read in conjunction with the recommended course timetable in Annex G – Sequencing and Timings.

Critical Training Resources	PO/EO
DWAN Laptop (each trainee)	001 - 006
DWAN drops (each trainee)	001 - 006
DRMIS software/portal (each trainee)	001, 003
Scientific Calculator (each trainee)	002
GLM software (each trainee)	002
BDCS trainer	002
Digital stability modelling software	002, 003
CPF Docking Video	002, 003
MS Project software	005

COURSE CAPACITY

5. Planning for the conduct of training is based on the guidelines in CFITES Vol 5 Annex D (Instructor to Learner Ratios). Justification can be found in Annex C.

a. Maximum – 12;

b. Minimum – 6; and

c. Optimum – 10.

MITE

6. Loading. The authority to load members onto this course lies with the NFS. NPTG is to be included as an INFO addressee on all loading messages.

7. Scheduling. It is to be scheduled in MITE by the NFS conducting the training and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends, and training system holidays.

8. Prerequisites. To be selected for this course the loading authority must ensure that trainees have the following prerequisites:

a. MAR TECH RQ-PO2 (ALUW); and

b. USQ Mechanical Supervisor (ALUX) or USQ Electrical Supervisor (ALUY).

9. Qualification. Successful completion of this course constitutes the primary means of attaining the MAR TECH RQ-PO1 (MITE 121848). Additionally, the following MITE Course ID will be granted: DRMIS Maintenance Manager Course - MITE 119098.

COURSE REPORTING

10. Upon passing this course, trainees shall receive a Course Report (CF 377) and Certificate of Military Achievement (CF 289) completed IAW DAOD 5031-9 "Course Reporting and Certificates" and NPTORD 4500-51 "Course Reports and Certification".

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

11. PLARs shall be conducted in accordance with NAVORD 4500-2 "RCN Prior Learning Assessment and Recognition", DAOD 5031-1 "Canadian Forces Military Equivalencies Program", and CFITES Vol 12 "Canadian Forces Military Equivalencies Programme: Prior Learning Assessment".

SPECIAL INSTRUCTIONS

12. PO 001 will be run as module 3 of the DRMIS Course: DRMIS Maintenance Manager. It will be run IAW the DRMIS Course QSP and trainees will be granted MITE 119098 on completion. It will be scheduled as the first part of the RQ-PO1 training.

13. The NCR Tour is to be arranged the first week of the course by the course staff. All trainees will need to provide their TIN for bookings. The students will be required to determine an "In progress Engineering Change during the first week of the course", this will be the subject for a presentation/PC to be based on the NCR visit.

14. SME lecturers are to be booked once course loading and dates are confirmed with as much notice as possible. Instructors will contact SMEs ahead of their lectures to confirm Teaching Points are covered off.

15. Lesson Objectives throughout this QSP are rated from Low to High priority for development in the standards paragraphs of each EO and PO. Criteria for rating is IAW the following:

a. Low: Impact to the trainees is low if this training is not completely developed. These LOs are relatively easy and frequently performed or theory-based where proficiency can be gleaned from the TP table, references, and/or experience;

b. Medium: LOs require further development than present in the TPt table / references. Instruction must be to a level where the trainees can accomplish the task competently; and

c. High: Impact to the trainees is high if this training is not effectively developed. These LOs are generally related to safety where retention is critical to minimize the likelihood of skill fade.

CHAPTER 2 ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

GENERAL

1. This is a pass/fail course. Trainees must successfully complete all POs to pass this course. Performance Checks (PCs) and Enabling Checks (ECs) are based on the Performance Objectives (POs) and Enabling Objectives (EOs) in Chapter 3 of this QSP. All PCs must be successfully completed to achieve the qualification.

2. Trainees will be developed and assessed using a combination of formative and summative assessment events. At a minimum, these will involve PCs, ECs, end of lesson confirmation, homework, assignments, and/or quizzes. See Annex K (Assessment Plan) for details.

3. Dress and deportment, ethics, conduct and behaviour demonstrated by the trainee shall be evaluated and assessed based on the standards laid out under Operation Honour and the LRH program. A trainee who displays inappropriate dress and deportment, ethics, and/or behaviour is considered to have demonstrated unsatisfactory conduct which may result in the convening of a Training Review Board (TRB).

ENABLING AND PERFORMANCE CHECKS

4. Enabling Checks (ECs). ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning to improve retention of course content. Formative assessments are designed to provide feedback to both instructional staff and trainees. Each EO must be formatively assessed prior to summative assessments.

5. In the event of an EC failure, the trainee will be counselled and given remedial training and/or additional homework, as appropriate. In the event of a consecutive EC failure, the trainee will be counselled IAW CFITES vol 6 with the unsatisfactory progress documented in their Trainee File.

6. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

7. PCs will be conducted after formative assessment has occurred and the trainee has had the opportunity to develop skills and apply knowledge related to the PO.

8. All PC failures and supplemental failures are to be handled IAW NPTORD 4500-54 “Training Review Board Process”. Supplemental PCs will normally be scheduled not less than 24 hours post counselling, depending on assessment resource availability.

RESPONSIBILITY FOR TESTING

9. Assessment of trainees is to be done IAW CFITES Vol. 7 “Evaluation of Leaners”.

10. NFS staff provides quality control by administering and scoring all ECs and PCs IAW the standards laid out in this QSP and the assessments developed by NTDC(P).

PROGRESS MONITORING

11. Trainee progress will be monitored using formative assessment tools to help the trainee and instructional staff recognize progress or lapses in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. Formative assessment is used during a phase of instruction to reinforce learning, so that content is retained.

12. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW TE SOs or SOPs and CFITES Vol 6 “Conduct of Instruction Programmes”.

13. Continuous monitoring of candidate progress is required to provide the following:

    a. Early warning of trainee difficulties;

    b. Feedback on the effectiveness of training;

    c. Information for any required TRBs; and

    d. Information to assist in completion of the CF 377 Course Report.

14. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs. A record of trainee counselling is to be maintained by the training establishment.

15. Instructional staff shall maintain individual trainee files, including the results of all:

    a. Trainee performance documentation (e.g. Weekly write-up, medical chits, dress inspection chits, conduct chits, etc.);

    b. Formative assessment material (e.g. ECs, practice periods, homework, assignments and quizzes, etc.); and

    c. Summative PCs.

TRAINING REVIEW BOARD

16. TRBs shall be convened and conducted IAW with NPTORD 4500-54 “Training Review Board Process”.

17. In the event of prolonged trainee absence, TE Staff shall determine, on a case by case basis, the necessity to convene a TRB.

COURSE REPORTS

18. Course reports (CF 377) shall be prepared and distributed IAW DAOD 5031-9 “Course Reporting and Certificates” and amplified in NPTORD 4500-51 “Course Reports and Certifications”.

19. Trainees will be graded pass or fail. The narrative will indicate the trainee’s strengths and weaknesses based on PC results to provide information to the Career Manager.

20. The Course Report narrative comments could include:

a. Ability to work with others;

b. Ability to work under pressure;

c. Written and verbal communication skills;

d. Organizational skills; and

e. Leadership ability.

21. Training limitations and / or incomplete training must be included in the course report.

CHAPTER 3 PERFORMANCE OBJECTIVES AND LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE CCA – NFS(P)

PO 001 Utilize DRMIS SAPGUI as a Maintenance Manager

1. This PO will be captured by the existing DRMIS Maintenance Manager Course (MITE 119098) IAW Mod 3 of the DRMIS Course QSP. MITE 119098 will be awarded on successful completion.

2. The DRMIS course incorporates task AFT0650 “Update the fitted equipment list (FEL)”.

PO 002 Manage Ship's Stability

1. Performance. Manage Ship's Stability (WT0070).

2. Conditions.

   a. Given:

      (1) References;

      (2) Workstation, with access to DWAN;

      (3) BDCS/GLM;

      (4) Assistance;

      (5) Supervision; and

      (6) Formulas.

   b. Denied: Nil.

   c. Environmental: Any condition on ship while at sea or alongside.

3. Standard. The MAR TECH manager shall manage ship's stability IAW the Manual of Trim and Stability by:

   a. (High) Assessing ship's stability (WT0040) to include:

      (1) Coordinating fuel schedules;

      (2) Managing amounts of liquid load;

      (3) Coordinating resource and equipment embarkation; and

      (4) Adjusting stability based on mission requirements to include the following:

         (a) Cargo;

         (b) Equipment; and

         (c) Personnel differences.

   b. (NT) Advising command on stability considerations.

   c. (High) Adjusting liquid and/or cargo load to maintain stability to include:

      (1) Identifying BDCS/GLM software requirements;

      (2) Inputting ship's data into BDCS/GLM; and

      (3) Identifying method of adjusting stability.

4. References. A1, and A2.

5. Task, Skill, Knowledge Numbers.WT0040.

6. Training Limitations. To be taught prior to PO 003 "Prepare vessel for docking/undocking" and PO 004 "Manage DC HQ" because it is the foundation for TPts covered in these POs.

PO 002 Manage Ship's Stability

EO 002.01 Assess Ship's Stability

1. Performance. Assess Ship's Stability.
2. Conditions.
   a. Given:
      (1) MT&S;
      (2) Formula sheets; and
      (3) Assistance.
   b. Denied: Nil
   c. Environmental: Classroom
3. Standard. The trainee shall assess ship's stability by:
   a. (Med) Coordinating fuel schedules.
   b. Managing amounts of liquid load to consider the following:
      (1) Effects of weight movement;
      (2) Effects of icing; and
      (3) Effects of grounding.
   c. Coordinating resource and equipment embarkation;
   d. Adjusting stability based on mission requirements to include calculating hydrostatic curves IAW platform specific MT&S for the following:
      (1) Deep departure;
      (2) Operation light;
      (3) Arrival; and
      (4) Docking.
4. Time Required.
   a. EO – 1200 mins to deliver training.
   b. EC – 115 mins (total for all trainees).
5. Remarks. Learning curve will may be steep especially with trainees who have limited previous exposure. Legacy E-Tech will have limited exposure.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
Coordinate fuel schedules and resource/equipment embarkation **LESSON OBJECTIVE: Describing stability and impact of embarkations (various):**			
1. Describing the purpose of shipboard stability, to include:	A1.	**Method / Media:** EDIP **Trg Environment:** Classroom	300

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
a. Transverse Stability; and b. Longitudinal Stability. 2. Defining the following stability terminology: a. Initial Stability. b. Large Angle Stability. c. Keel. d. Force. e. Center of Gravity, to include: i. Vertical Center of Gravity (VCG or KG); ii. Longitudinal Center of Gravity (LCG); and iii. Transverse Center of Gravity (TCG). f. Buoyancy, to include: i. Buoyancy volume; ii. Buoyant volume; iii. Reserve of Buoyancy; iv. Force of Buoyancy; and v. Centre of Buoyancy. g. Freeboard. h. Moment. i. Couple. j. List. k. Heel. l. Trim. m. Metacentre. n. Metacentric Height. o. Period of Roll, to include: i. Stiff Ship; and ii. Tender Ship. 3. Explaining conditions of initial stability, to include: a. Positive Stability. b. Negative Stability. c. Neutral.		**Activity Details:** Instructor will explain and demonstrate the different stability Terminology. **Trg Resources:** • Formula sheet; and • Lesson plan.	
LESSON OBJECTIVE: Explaining the measures of stability, to include:			
1. Metacentric Height. 2. Curve of Static Stability. 3. Righting Arms. 4. Range of stability. 5. Angles of stability, to include: a. Angle of list.	A1.	**Method / Media:** EDIP **Trg Environment:** Classroom **Activity Details:** Instructor will explain and demonstrate different stability measures.	200

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
b. Angle of deck immersion. c. Angle of vanishing stability. d. Angle of Maximum stability. 6. Cross curves of stability.		**Trg Resources:** • Formula sheet; and • Lesson plan.	
Manage amounts of liquid load **LESSON OBJECTIVE: Explaining the effects of a weight movement on stability, to include:**			
1. Vertical Weight Shift. 2. Transverse Weight Shift. 3. Combined Weight Shift. 4. Weight addition or removal.	A1.	**Method / Media:** EDIP **Trg Environment:** Classroom **Activity Details:** Instructor will explain and demonstrate the effects of a weight movement on shipboard stability. **Trg Resources:** • Formula sheet; and • Lesson plan.	200
LESSON OBJECTIVE: Explaining fluid related stability phenomenon's, to include:			
1. Free Surface Effect (FSE), to include pocketing 2. Free Communication Effect (FCE). 3. List. 4. Loll.	A1.	**Method / Media:** EDIP **Trg Environment:** Classroom **Activity Details:** Instructor will explain and demonstrate fluid related stability phenomenon. **Trg Resources:** • Formula sheet; and • Lesson plan.	100
LESSON OBJECTIVE: Explaining the effects of icing in regard to stability, to include:			
1. Icing impacts to the following: a. Stability. b. List and Trim. c. Freeboard. d. Maneuvering and Control. e. Equipment. f. Safety. 2. Factors that affect the accumulation of ice, to include: a. Air and Sea Temperatures. b. Wind Speed and Direction. c. Displacement and Freeboard. d. Sea State. e. Precipitation. f. Time of Day. 3. Actions taken to counter icing.	A1.	**Method / Media:** EDIP **Trg Environment:** Classroom **Activity Details:** Instructor will explain and demonstrate the effects of icing in regard to stability. **Trg Resources:** • Formula sheet; and • Lesson plan.	50
LESSON OBJECTIVE: Explaining the effects of grounding, to include:			
1. Effects on stability.	A1.	**Method / Media:** EDIP	50

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
2. Freeing a Grounded Vessel.		**Trg Environment:** Classroom **Activity Details:** Instructor will explain and demonstrate the effects of grounding. **Trg Resources:** • Formula sheet; and • Lesson plan.	
Adjust stability to include calculating hydrostatic curves **LESSON OBJECTIVE: Describing the MT&S to include:**			
1. Describing the purpose of an Inclining Experiment. 2. Explaining the different sections to draw information from for stability calculations, including general information: a. Vessel particulars. b. The different operational conditions, to include: i. Deep departure; ii. Operational light; iii. Arrival; and iv. Docking. c. Damaged stability and intact stability. d. Tank particulars. e. Watertight compartment particulars. f. Icing conditions. g. Hydrostatic curves/data to include: i. Displacement (Δ) (Salt water); ii. Tons Per Inch / Tonnes Per Centimetre (TPI / TPC) Immersion; iii. Longitudinal Centre of Flotation (LCF); iv. Moment to Change Trim by 1” / 1cm (MCT); v. Height of Transverse Metacentre (KM, KMT, or KML). vi. Vertical Centre of Buoyancy (VCB); and vii. Longitudinal Centre of Buoyancy (LCB).	A1.	**Method / Media:** EDIP **Trg Environment:** Classroom **Activity Details:** Instructor will explain and demonstrate using the MT&S. **Trg Resources:** • Formula sheet; • 1 x MT&S per trainee; and • 1 x MT&S per instructor.	300

7. Test Details. EO 002.01 will be tested in EC 002.01 and PC 002 IAW table below and Annex K.

EC 002.01: Assess Ship's Stability

EC Time: 100 mins + 15min debrief

ASSESSMENT ACTIVITY: The trainee will be given a two-part written exam. The trainee will be required to define different stability terminology for part one, then produce information from the MT&S based on given scenarios for part two. (I.e. vessel has 6 cm of ice on the port foc'sle, how much weight is this up high?).

Pass/fail criteria: 60% pass mark

ASSESSMENT RESOURCES:

- Exam;
- Formula sheet; and
- Manual of Trim and Stability.

IMPACT OF FAILURE: Remedial Training / Re-test.

PO 002 ***Manage Ship's Stability***

EO 002.02 **Adjust liquid or cargo load to maintain stability**

1. Performance. Adjust liquid or cargo load to maintain stability.
2. Conditions.
   a. Given:
      (1) BDCS work station;
      (2) Stand-alone GLM;
      (3) Scenarios;
      (4) Example pictures;
      (5) Example videos;
      (6) Digital Stability Modelling software; and
      (7) MT&S.
   b. Denied: Assistance.
   c. Environmental: Classroom/PCOT/BDCS trainer.
3. Standard. The trainee shall adjust liquid or cargo load to maintain stability IAW the applicable platform's MT&S by:
   a. (Med) Identifying BDCS/GLM requirements to include the following:
      (1) User levels;
      (2) Login information;
      (3) BDCS (IPMS) Stability Page inputs:
         (a) Tanks (auto);
         (b) Fuel report (auto);
         (c) Compartments;
         (d) Draught;
         (e) List;
         (f) Trim; and
         (g) Displacement;
      (4) Differences of PC based GLM (stand-alone) on various platforms.
   b. (High) Inputting ship's data into BDCS/GLM to include:
      (1) Identifying data types to include the following:
         (a) Bilge load;
         (b) Unusual loading;
         (c) Flooding;

(d) Icing;

(e) List;

(f) Trim;

(2) Identifying data outputs;

(3) Identifying automatic adjustments based on data input;

(4) Manually adjusting inputs based on data; and

(5) Saving files.

c. (Med) Identifying methods of adjusting stability to include:

(1) Prioritizing adjustment sequence;

(2) Identifying secondary effects to the ship during and post-adjustment; and

(3) Identifying risk factors during and post-adjustment.

4. Time Required.

a. EO – 430 mins to deliver training;

b. EC – 120 (total for all trainees); and

c. PC – 200 minutes (total for all trainees).

5. Remarks. Less time will be required for EC 002.02 back-briefs/debriefs if multiple assessors are available.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Identify BDCS/GLM software features			
1. User levels. 2. Login information. 3. BDCS (IPMS) Stability Page: a. Tanks (auto) b. Fuel report (auto) c. Compartments d. Draught e. List f. Trim g. Displacement. 4. Differences of PC based GLM (stand-alone) on various platforms.	A1, A2.	**Method / Media:** F2F **Trg Environment:** PCOT/BDCS Trainer **Activity Details:** Instructor to identify the various features of the BDCS/GLM while trainees follow along on their workstations. **Trg Resources:** • BDCS workstation for each trainee and one with projector for instructor • Stand-alone GLM for comparison demonstration	30
LESSON OBJECTIVE: Input ship's data into BDCS/GLM			
1. Identify data types: a. Bilge load	A1, A2.	**Method / Media:** EDIP **Trg Environment:**	50 (ED) 150 (IP)

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
b. Unusual loading c. Flooding d. Icing e. List f. Trim. 2. Identify data outputs. 3. Identify automatic adjustments based on data input. 4. Save files.		PCOTT/BDCS trainer **Activity Details:** Instructor to explain and demonstrate how to input ship's data. Trainees to experiment with data to see how the BDCS/GLM/ship reacts to the changes. **Trg Resources:** • BDCS workstation for each trainee and one with projector for instructor • Scenarios	
LESSON OBJECTIVE: Identify different methods of adjusting stability			
1. Methods of adjustment: a. Ballasting b. Fuel management c. Freshwater management d. B&G water management e. Bilge management f. Stores/load management. 2. Prioritizing adjustment sequence: a. Trim b. List c. Draught d. Preferred medium (eg. water before fuel). 3. Identifying secondary effects: a. Increased list / trim b. FSE c. FCE d. Effects on DC systems. 4. Identifying risk factors: a. Contamination b. Safety of personnel c. Safety of equipment d. Environmental considerations.	A1, A2.	**Method / Media:** GD **Trg Environment:** Classroom **Activity Details:** Instructor will ask open questions about the various methods of adjusting the ship's stability. Trainees will discuss. Instructor to input added questions as required to ensure all TPts are covered off. Instructors to show examples of poor stability. **Trg Resources:** example videos, pictures, etc.	200

7. Test Details. EO 002.02 will be tested in EC 002.02 and PC 002 IAW table below and Annex K.

EC 002.02: Adjust liquid or cargo load to maintain ship's stability

EC Time: 20 mins (concurrent) + 10 min back-brief with assessor debrief (individual) [Total of 120 mins]

ASSESSMENT ACTIVITY: Trainees will be given a random set of stability data to input into BDCS/GLM, in the correct sequence, to stabilize the platform. Trainees will verbally brief MSEO (assessor) on their stability plan.
Pass/fail criteria: Trainees must achieve a SAT on all checklist items to be successful with this assessment.

ASSESSMENT RESOURCES:

- BDCS/GLM
- Stability data
- Scenario

IMPACT OF FAILURE: Remedial Training / Re-test

MAR TECH RQ-PO1 – EC 002.02 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Enter data into BDCS/GLM			
Accurate data entered			
Data entry complete			
2. Brief stability plan to MSEO (Assessor)			
Based on BDCS/GLM end state			
Prioritize sequence of adjustments based on current situation			
Identify secondary effects ✓ What will happen to the ship and systems with each adjustment (list, trim, hull, emergency fitted systems, etc.)			
Identify risk factors during and post-adjustment ✓ Environmental considerations ✓ Safety considerations ✓ Personnel considerations ✓ Resource considerations			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 002: Manage Ship's Stability

PC Time: 200 mins

ASSESSMENT ACTIVITY: The trainee will be given a stability scenario and will conduct stability adjustments using both BDCS/GLM and the Manual of Trim and Stability. The trainee will be required to develop a plan to stabilize the ship to bring it back to a "safe at sea" condition. The trainee will write recommendations to MSEO (format will not matter). The MSEO (assessor) shall input adjustments into BDCS to confirm they are within margin of error (+/- 3° port/stbd; draught +/- .25m LCF amidships). Trainee will have a maximum of two attempts within the allotted time limit.
Pass/fail criteria: Stability is confirmed by BDCS to be +/- 3° port/stbd; draught +/- .25m LCF amidships based on trainee's recommendations.
Scenario to encompass:

- Ballast and storage condition (i.e. tanks full/empty);
- Fuel state; and
- Current ship's list (port/stbd) and trim (fwd/aft).

ASSESSMENT RESOURCES:

- BDCS/GLM;
- Manual of Trim and Stability;
- Scenario; and
- Formula Sheet.

IMPACT OF FAILURE: Remedial Training / Re-test with new scenario / TRB after unsuccessful retest.

PO 003 Prepare a Vessel for Docking / Undocking

1. Performance. Prepare a Vessel for Docking / Undocking.
2. Conditions.
   a. Given:
      (1) Ship;
      (2) Assistance;
      (3) References.
   b. Denied: Nil.
   c. Environmental: Ship alongside, day or night.
3. Standard. The MAR TECH Manager shall prepare a vessel for docking / undocking IAW direction from the Dock Master by:
   a. (High) Managing docking preparations (AKT1190) to include:
      (1) Preparing for routine docking;
      (2) Performing post-docking verifications; and
      (3) Preparing for unscheduled docking; and
   b. (Med) Managing undocking preparations (GT0040).
4. References. A1, A17, A20, D1, and D2.
5. Task Numbers. AKT1190; GT0040.
6. Training Limitations. Nil.

PO 003 ***Prepare Vessel for Docking/Undocking***

EO 003.01 **Manage Docking Preparations**

1. Performance. Manage Docking Preparations (AKT1190).
2. Conditions.
   a. Given:
      (1) DWAN workstation;
      (2) CF 523 "Certificate of Docking and Undocking";
      (3) Docking POC list;
      (4) Docking reference list;
      (5) Dock and Ship (for tour); and
      (6) Reference.
   b. Denied: Nil.
   c. Environmental: Classroom.
3. Standard. The trainee shall manage docking preparations (AKT1190) IAW CF 523 Docking/Undocking Certificate by:
   a. (Med) Identifying the components of a docking plan.
   b. (High) Preparing for routine docking to include:
      (1) Identifying pre-docking milestones;
      (2) Identifying docking milestones; and
      (3) Identifying docking failures.
   c. (Med) Performing post-docking verifications.
   d. (High) Preparing for unscheduled docking to include:
      (1) Identifying occasions for unscheduled docking;
      (2) Liaising with POCs;
      (3) Calculating docking stability; and
      (4) Completing unscheduled docking.
4. Time Required.
   a. EO – 950 mins to deliver training;
   b. TTT – 60 mins transport for field trip; and
   c. EC – 60 mins (total for all trainees).
5. Remarks.
   a. Dock Tour: Instructor will need to confirm the dock tour well ahead of time to ensure that all candidates have the proper requirements from the Dock

Master/Officer. Tour should be booked for a dock which is currently being used, with HMC vessel if possible.

b. SME Lecture: Instructor to confirm SME lecturer ahead of time. They should inform the SME of required topics for the lecture.

c. Possible references for developing this EO:

(1) CFTO D-03-002-008/SG-011 - Requirements for Docking/Undocking of CF Vessels.

(2) Docking Instructions US Navy S9086-7G-STM-010/CH-997.

(3) C-09-153-003/TS-000, Explosive Safety Manual, Volume 3.

(4) Naval Engineering Manual – Part 6, Section 4 Docking.

(5) Docking plan: HFX-D28-589-000-13 (drawing number).

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Define docking terminology			
1. Docking terms (in MT&S) a. Sueing. b. Lift-off. 2. Types of docks a. Graving/Basin dock. b. Floating dry dock. c. Marine railway. d. Synchro-lift. 3. Types of blocks and shores a. Crib. b. Double crib. c. Treble crib. d. Components of keel blocks: i. Caps; ii. Blocks; iii. Ground block; iv. Securing dogs; v. Wedges; and vi. Strapping. e. Shores: i. Bottom shore; and ii. Breast shore. 4. Types of caissons: a. Floating. b. Sliding. c. Rolling. 5. Docking occasions.	A1, D1, D2 Art. 1.3 (docks).	**Method / Media:** Field trip (Dock tour); F2F with visual aids **Trg Environment:** Dry Dock for tour; classroom **Activity Details:** Trainees will get a dock tour during which the instructor/dock master/staff will highlight key dock components. After tour, instructor will show remain items in the classroom **Trg Resources:** • Dry-dock with components • Dock Master (SME instructor) • Photos/models of components	*60 (for transport)* 100 (tour) 50 (F2F)

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Identify the components a docking plan			
1. Identify information recorded on CF 523. 2. Describe responsibilities of personnel involved in docking operations: a. Dock Master. b. Docking Officer. c. QAWC. d. Project manager. e. Pilot. f. Ships Staff: i. CO; ii. XO; and iii. HODs/CHODs. 3. Identify location of docking plan (Captain's Ship's Books). 4. Identify information found on a Ship's Docking Plan. 5. Explain Docking Certificate. 6. Explain Change of Weight in Dock form. 7. Identify pre-docking arrangements. 8. Explain dock preparation requirements. 9. Identify the damage control requirements for docking. 10. Identify machinery state requirements for docking. 11. Identify liquid load state requirements for docking. 12. Explain requirement to take breakage of ship measurement prior to docking. 13. Identify regulations concerning docking with ammunition on board.	A17, D1.	**Method / Media:** F2F; SME lecture **Trg Environment:** Classroom **Activity Details:** Instructor will lead the trainees through a routine docking plan including all components listed in the TPts. Dock Master/Docking officer SME will discuss how a docking happens from their perspective and answer questions from the trainees. **Trg Resources:** • Dock Master SME Lecturer • Pictures of components and example dockings/dock preps; docking failures • CF 523 • Reference handout • Docking Master Plan (example)	400 (F2F) 100 (SME)
LESSON OBJECTIVE: Prepare for a routine docking			
1. Identify pre-docking milestones: a. De-ammunition. b. De-fuel. c. De-store.	A17.	**Method / Media:** EDI **Trg Environment:** Classroom **Activity Details:**	100 (E) 60 (DI)

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
d. Comply with stability requirements IAW Dock Master instructions: i. Trim; ii. List; and iii. Draught. e. Comply with engineering requirements (equipment, DC conditions, etc.). f. Comply with DC requirements (day of docking). g. Complete dock walk (day of docking). h. Position personnel accordingly for cold move (sentries, etc.). 2. Identify docking milestones: a. Cold move to dry dock. b. Transfer custody to RF. c. 1' prior to sueing (divers block alignment). d. 1' after sueing (divers confirm block fit). e. Isolate suctions/discharges (depends on equipment required). 3. Identify docking failures: a. Not isolating systems properly. b. Mechanical failures. c. Bad block positioning. d. Bad ship position. e. Block failure. f. Improper loading.		Instructor will explain routine docking preparations including all milestones listed in the TPts. Instructor will then demonstrate the docking milestones using stability digital modelling software and the CPF docking video. Instructor will show pictures/visual aids of docking failures and explain / demonstrate repercussions of failures using the software. Trainees will take turns using the stability modelling software to see how different platforms will react in different situations. **Trg Resources:** • CPF docking video "Preparing Ships for Docking" • Stability digital modelling software • Pictures of docking failures	50 (demo)
LESSON OBJECTIVE: Perform post-docking verifications			
1. Maintain weight control log. 2. Survey the ship's hull. 3. Manage ships service requirements: a. Fire fighting. b. Mag flooding. c. Hotel services. d. Overboard discharges.	A17.	**Method / Media:** GD, pictures **Trg Environment:** Classroom **Activity Details:** Instructor will ask questions to guide the trainees through a discussion of the TPts. Instructor to confirm all TPts are covered by the discussion and show pictures of the hull,	50

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
4. Coordinate procedures & requirements for turning shafts and equipment while in dock.		discharges, etc. while a ship is docked. Instructor to show an example of a weight control log **Trg Resources:** • Pictures • Weight control Log example	
LESSON OBJECTIVE: Prepare for unscheduled docking			
1. Describe occasions for unscheduled docking: a. Grounding. b. Collision. c. Allision. d. Mechanical failure. e. Structural failure. f. Battle damage. g. Fouling. 2. Liaise with POCs: a. NavArc/FTA. b. F4Eng. c. FMF. 3. Describe docking stability calculations for unscheduled docking: a. Fuel. b. Personnel. c. Stores. 4. Carry out unscheduled docking: a. Modified machinery considerations. b. Modified DC considerations c. Cooling requirements. d. Discharge/Intake.	A17.	**Method / Media:** GD, demo **Trg Environment:** Classroom **Activity Details:** Instructor will ask questions to guide the trainees through a discussion of the TPts. Instructor to confirm all TPts are covered by the discussion. Instructor will show visual aids comparing unscheduled docking modifications versus scheduled docking configurations. **Trg Resources:** • POC list handout • Pictures of unscheduled docking modifications	40

7. Test Details. EO 003.01 will be tested in EC 003.01 and PC 003 IAW table below and Annex K.

EC 003.01: Manage docking preparations

EC Time: 50 mins + 10 min debrief

ASSESSMENT ACTIVITY: Trainees will be given a written exam for docking a ship. The written exam shall contain at least 20 questions related to docking a ship including the requirement to fill in applicable portions of a CF 523 Certificate of Docking and Undocking. Trainees will be required to reference their answers.
Pass/fail criteria: 80%.

ASSESSMENT RESOURCES:

- Workstation with DWAN access;
- CF 523 Certificate of Docking and Undocking; and
- Information for CF 523.

IMPACT OF FAILURE: Remedial Training / Re-test.

PO 003 ***Prepare Vessel for Docking/Undocking***

EO 003.02 **Manage Undocking Preparations**

1. Performance. Manage Undocking Preparations (GT0040).
2. Conditions.
   a. Given: CF 523 Certificate of Docking and Undocking.
   b. Denied: References.
   c. Environmental: Classroom.
3. Standard. (Med) The trainee shall manage undocking preparations (GT0040) IAW CF 523 Docking/Undocking Certificate by:
   a. Identifying pre-flooding requirements.
   b. Determining required stability condition of vessel for undocking.
   c. Completing the undocking certificate.
   d. Identifying who is responsible for undocking stability requirements.
   e. Identifying procedures to float out vessel.
4. Time Required.
   a. EO – 60 mins to deliver training; and
   b. PC – 70 minutes (total for all trainees).
5. Remarks. No practical assessment for this phase because of resource requirements and time to manage undocking procedures as performed on the job.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Identify undocking arrangements			
1. Identify pre-flooding requirements to include: a. Hull integrity. b. Plugs and blanks removed. c. Dock walk. 2. Describe required stability condition of vessel for undocking. 3. Explain the undocking certificate. 4. Identify who is responsible for undocking stability calculations (Dock Master). 5. Describe procedures to float out vessel.	D1, A20.	**Method / Media**: GD; Demo **Trg Environment:** Classroom **Activity Details:** Instructor will ask questions to guide the trainees through a discussion of the TPts. Instructor to confirm all TPts are covered by the discussion and demonstrate using a digital stability/docking modelling software how undocking looks (proper and improper). Trainees will be given a 10-question assignment to complete on their own time to assess their comprehension. It	50 10 (debrief)

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
		will be reviewed in class by the instructor. **Trg Resources:** • digital stability/docking modelling software (with projector)	

7. Test Details. EO 003.02 will be tested in PC 003 IAW table below and Annex K.

PC 003: Prepare vessel for docking/undocking

PC Time: 60 mins + 10 min debrief

ASSESSMENT ACTIVITY: Trainees will be given a written exam for docking and undocking a ship. The written exam shall contain at least 30 questions related to docking and undocking a ship including the requirement to fill in applicable portions of a CF 523 Certificate of Docking and Undocking. Trainees will not be allowed references.
Pass/fail criteria: 70%.

ASSESSMENT RESOURCES
- CF 523 Certificate of Docking and Undocking template; and
- Information for CF523.

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after unsuccessful retest.

PO 004 Coordinate Departmental Readiness Requirements

1. Performance. Coordinate Departmental Readiness Requirements.

2. Conditions.

   a. Given:

      (1) References;

      (2) Assistance;

      (3) Operational ship; and

      (4) Computer station with DWAN access.

   b. Denied: Nil.

   c. Environmental: Any conditions, day, or night, at sea or alongside.

3. Standard. The MAR TECH manager shall independently coordinate the departmental readiness requirements IAW refs by:

   a. (Med) Applying readiness requirement policies to include:

      (1) Determining organization's roles and accountability;

      (2) Managing materiel certification program (AFT0100); and

      (3) Enforcing configuration management.

   b. (Med) Managing readiness rhythm to include the following:

      (1) Extended readiness;

      (2) Restricted readiness;

      (3) Normal readiness to include force generation;

      (4) High readiness to include force employment; and

      (5) Tiered readiness program.

4. References. A9, A10, A11, A12, A17, A15, A7, A18, and A9.

5. Task, Skill, Knowledge Numbers. AFT0100, AFT0170, AFT0090, AKT1240, ZT0070, AET1010, and AFT0210.

6. Training Limitations. Scheduling: The timing of this PO should occur prior to the planned field trip to the NCR to allow the trainees to capitalize on the material contained within.

PO 004 ***Coordinate Departmental Readiness Requirements***

EO 004.01 Apply readiness requirement policies.

1. Performance. Apply readiness requirement policies.

2. Conditions.

   a. Given:

      (1) References;

      (2) Assistance; and

      (3) Computer station with DWAN access.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall independently apply readiness requirement policies IAW refs by:

   a. (Low) Determining organization's roles and accountability to include the following:

      (1) National Defence Headquarters components; and

      (2) Local Maritime components.

   b. (Low) Managing materiel certification program (AFT0100) to include:

      (1) Assuring the materiel state of naval systems and equipment by:

         (a) Assuring compliance with applicable legislation and approved standards;

         (b) Conforming to MBS; and

         (c) Facilitating risk management.

      (2) Reporting deficiencies through OPDEFS.

   c. (Low) Enforcing configuration management by:

      (1) Configuring item selection;

      (2) Accepting new equipment to include the following:

         (a) Technical evaluation; and

         (b) Engineering changes (EC).

      (3) Controlling configuration of fitted items to include updating FLOC (Functional location of Item).

4. Time Required.

   a. EO – 550 mins to deliver training; and

   b. EC – 50 mins (total for all trainees).

5. Remarks. Nil.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Describing organization's roles and accountability to include:			
1. National Defence Headquarters components to include: a. Assistant Deputy Minister (Materiel). b. Chief of the Maritime Staff. c. Director General Maritime Equipment Program Management. d. Director Maritime Management and Support. e. Director Maritime Ship Support. f. Director Maritime Equipment Program Management (Submarines). g. Class Desks. h. Director Maritime Procurement. i. Other Divisions within the Assistant Deputy Minister (Materiel) Organization. 2. Local Maritime components to include: a. Formation Commander. b. J4/N4 Assistant Chief of Staff Naval Engineering and Maintenance. c. J42/N42 Formation Technical Authority. d. J37/N37 Engineering Operations. e. Fleet Technical Officer. f. Canadian Forces Naval Training Establishments. g. J6/N6 Assistant Chief of Staff Communications and Information Systems. h. Fleet Maintenance Facility. i. Ship Personnel. j. Commanding Officer. k. Executive Officer. l. Engineering Head of Department. m. Senior Maintainer and Section Head. n. Departmental Maintenance Coordinator.	A14 PART 4.	**Method / Media:** IL **Trg Environment:** Classroom **Activity Details:** Instructor at first will give a lecture identifying roles and accountabilities. The trainees will then work as a group to build a favourites list of links to the different organizations that they will be able to make a takeaway. Instructor to confirm that trainees have found the correct references. **Trg Resources:** • DWAN workstation, and • Link support job aid.	100 mins for lecture 50 mins for activity 150 mins total

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
o. Maintenance Supervisor. p. Maintainers.			
LESSON OBJECTIVE: Managing materiel certification program (AFT0100) by:			
1. Assuring the materiel state of naval systems and equipment by: a. Assuring compliance with applicable legislation and approved standards. b. Conforming to MBS. c. Facilitating risk management; and d. Reporting deficiencies by: i. OPDEFS; and ii. Notification of Defect (ND).	A9, A10, A12, A17, A15.	**Method / Media: GS, IL** **Trg Environment:** Classroom. Activity Details: Guest lecture from Fleet Technical Authority to speak to trainees focusing on Naval Material Assurance and Naval Material Regulations. Interactive lesson to consist of previously submitted OPDEFS to N37 to be used as examples to discuss issues with submissions. Second interactive lesson to focus on the role of each OPDEF addressee. Third interactive lesson to focus on risk management policy. Trainees will be required to analyze risk assessments submitted to FTA. Fourth interactive lesson to focus on adherence to MBS and effect on ship's operation. **Trg Resources:** • OPDEF examples; and • Guest lecturer.	50 mins for guest speaker 250 min for IL 300 min total
LESSON OBJECTIVE: Enforcing configuration management by**:**			
1. Configuration item selection by: a. Describing the selection process to include: i. X ship trials and role. 2. Accepting new equipment to include: a. Technical evaluation. b. Engineering changes (EC). c. Controlling of configuration fitted items including FLOC.	A14 PART 4.	**Method / Media:** IL/teleconference. **Trg Environment:** Classroom**.** **Activity Details:** Lesson to consist of the instructor covering all of the indicated teaching points. Practical examples of trials that were completed on the X ship and have been implemented in the fleet. Lesson followed up with a teleconference with NFR representative to discuss current requirements in trials. **Trg Resources:** • Lesson PPt • Telecommunication apparatus.	50 mins for lesson 50 min for teleconference 100 mins total.

7. Test Details. EO 004.01 will be tested in EC 004.01 and PC 004 IAW table below and Annex K.

EC 004.01: Apply readiness requirement policies

EC Time: 40 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: Trainee will receive a scenario based on a real life event, in which there is a failure of equipment that brings the ship below MBS. The trainee will be required to conduct a risk assessment and identify the subsequent OPDEF's priority.
Pass/fail criteria: 80% pass mark.

ASSESSMENT RESOURCES:

- Scenario;
- Access to DWAN;
- OPDEF template; and
- References.

IMPACT OF FAILURE: Remedial Training / Re-test.

RQ PO1 MAR TECH Manager – EC 004.01 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve an 80% SAT on the checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Apply readiness requirement policies			
Determines MBS compliance			
Correctly complete risk assessment matrix			
Determined mitigation factors			
Completed mitigated risk assessment matrix			
Assigns correct priority to OPDEF			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 004 Coordinate Departmental Readiness Requirements

EO 004.02 Manage readiness rhythm

1. Performance. Manage readiness rhythm.
2. Conditions.
   a. Given:
      (1) References;
      (2) Assistance; and
      (3) DWAN workstation.
   b. Denied: Nil.
   c. Environmental: Classroom.
3. Standard. The trainee shall independently manage readiness rhythm IAW refs by:
   a. (Low) Determining readiness levels to include the following:
      (1) Extended readiness;
      (2) Restricted readiness;
      (3) Normal readiness to include force generation;
      (4) High readiness to include force employment; and
      (5) Tiered readiness program.
   b. (Low) Managing short Work Periods (AFT0210) to include work period planning processes.
   c. (Low) Co-ordinating departmental combat readiness requirements (AET1010).
   d. (Low) Managing maintenance profiles to include:
      (1) Tracking maintenance schedules; and
      (2) Liaising with other agencies.
   e. (Low) Determining notice for power to include (ZT0070) the following:
      (1) Machinery state requirements; and
      (2) Types of Notice for Power.
4. Time Required.
   a. EO – 350 mins to deliver training;
   b. EC – 50 mins (total for all trainees); and
   c. PC – 215 mins (total for all trainees).
5. Remarks. Nil.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Describing readiness levels to include:			
1. Extended readiness. 2. Restricted readiness to include tiered readiness program. 3. Normal readiness to include force generation. 4. High readiness to include force employment. and 5. Tiered readiness program 6. Notice for power.	A7.	**Method / Media:** GL/IL. **Trg Environment:** Classroom. **Activity Details:** Trainees to have reviewed CFCD 129 prior to arrival of guest lecturer. Guest lecturer from N37 to provide an overview of the different readiness levels. Focus on the tiered readiness program and responsibilities of the MAR TECH manager. **Trg Resources:** • Guest lecturer; and • Access to CFCD 129.	50 min
LESSON OBJECTIVE: Managing short Work Periods by:			
1. Applying work period planning process to include: i. Project initiation; ii. Project definition and planning; iii. Project execution and monitoring; and iv. Project closure.	A18 para 4.	**Method / Media:** GL. **Trg Environment:** Classroom**.** **Activity Details:** Ship Service Officer to provide guest lecture on the work period planning process. Work periodization to be discussed based on J37 assignments. **Trg Resources:** • Guest lecturer; • Work period planning process para 4 table as performance support.	100 mins
LESSON OBJECTIVE: Co-ordinating departmental combat readiness requirements			
1. Overview to include: a. Purpose. b. Relevance. c. Organization. d. Responsibilities. e. Guiding Principles. f. Serial Construct. g. Capability Streams. h. Assessment. i. Reporting. j. Amendments. k. Verification. and l. Collective training. 2. Application of the STG to meet serial requirements. To include:	A13, STG, A19.	**Method / Media:** GL. **Trg Environment:** Classroom. **Activity Details:** Guest lecturer from STG to present an overview of the process of achieving departmental CRRs. Trainees required to read through references prior to lesson and prepare questions to ask guest lecturer. **Trg Resources:** • STG guest lecturer; • DWAN workstation; • References; and • Example of CRR tracking sheet.	75 mins

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
a. Identifying required departmental serials. b. Scheduling serials. c. Material requirements. d. Personnel requirements. e. Personnel tracking.			
LESSON OBJECTIVE: Managing maintenance profiles by:			
1. Tracking maintenance schedule to include: a. Planned maintenance; b. Corrective maintenance; and c. Tracking reports via DRMIS COE (Bob J report). 2. Liaising with other agencies to include: a. CoC; b. Internal departments; and c. External agencies. 3. Quality assurance overview (FMF).	A14.	**Method / Media:** IL. **Trg Environment:** Classroom. **Activity Details:** Interactive lecture to be taught by instructor to cover all pertinent teaching points. Trainees will be assigned homework in which they will be required to read Part 6 of NaMMS and answer questions on a take-home assignment. **Trg Resources:** • Bob J example • Homework assignment	75 mins
4. Determining notice for power to include (ZT0070) the following: a. Machinery state requirements; and b. Types of Notice for Power: immediate, 4 hours, 8 hours, 15 minute, 48 hours, extended.	A19.	**Method / Media:** IL. **Trg Environment:** Classroom. **Activity Details:** Interactive lecture to be taught by instructor to cover all pertinent teaching points.	50 mins

7. Test Details. EO 004.02 will be tested in EC 004.02 and PC 004 IAW table below and Annex K.

EC 004.02: Manage Readiness Rhythm

EC Time: 40 mins (concurrent) + 10 min debrief

ASSESSMENT ACTIVITY: Trainees will receive a detailed letter from Sea Training staff, which identifies the ship's CRRs. Trainee will be tested one of the following CRRs: 2W09, 10W02, 10w03, 10W04, 10W05. They will be given the serial by the instructor and be required to develop the plan to execute the serial on board ship.
Pass/fail criteria: Passing 70% of EC checklist.

ASSESSMENT RESOURCES:
- Sea Training staff letter;

- Access to references; and
- CRR tracking sheet.

IMPACT OF FAILURE: Remedial Training.

RQ PO1 MAR TECH Manager – EC 004.02 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a 70% SATs on the checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	SAT	UNSAT	COMMENTS
1. Managing Readiness Rhythm			
Identify required CRRs from Sea Training letter and CRR tracking sheet.			
2. Build training plan to meet CRR training requirements			
Use CRR as reference			
List personnel required			
List material required			
List time required			
List safety considerations			
Define required setup to execute serial			
Identify location of setups			
Identify time required for each setup			
Identify plan to move groups between setups			
Designate a SME at each station.			
Designate a safety personnel for each station			
Clearly define required tasks and expected level of performance			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 004: Coordinate Departmental Readiness Requirements

PC Time: 200 mins concurrent + 15 min debrief.

ASSESSMENT ACTIVITY: Trainee will be provided a scenario chosen at random which will describe a ship's current point in a 5-year cycle. Using the information, and applicable references, the trainee will draft a briefing note to the CO. The briefing note will include the following:

- A detailed description of the current status of the TRP;
- Upcoming TRP milestones;
- Risks and deficiencies in critical areas;
- Mitigation recommendations; and
- Summary.

Pass/fail criteria: Passing 70% of PC checklist.

ASSESSMENT RESOURCES:

- Randomly created scenario;
- References;
- DWAN workstation; and
- Briefing note template.

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt

RQ PO1 MAR TECH Manager – PC 004 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a 70% SAT on the remainder of the checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial training, re-test / TRB after unsuccessful retest.

ITEM	SAT	UNSAT	COMMENTS
1. Coordinate Departmental Readiness Requirements			
Trainee follows briefing note template			
Maximum 2 pages			
Correctly identifies ship's current state in 5 year cycle			
Identifies relevant milestones			
Describes pertinent risks and deficiencies in relation to current state in 5 year cycle			
Mitigation recommendations are realistic and achievable based on scenario			
Summary is concise			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 005 Manage DND Projects

1. Performance: Manage DND projects.

2. Conditions:

   a. Given:

      (1) References;

      (2) IT resources (DWAN Access); and

      (3) Assistance.

   b. Denied: Nil.

   c. Environmental: Any conditions, at sea or ashore.

3. Standard. The MAR TECH Manager shall manage small DND projects IAW applicable references by:

   a. (High) Initiating a project proposal to include:

      (1) Completing a feasibility study;

      (2) Initiating a project charter;

      (3) Establishing TORs for project team members;

      (4) Coordinating facility requirements for contractors (AFT1110); and

      (5) Establishing project timeline(s).

   b. (High) Creating a project plan to include:

      (1) Identifying the critical path (the central focus and end-goal of the project);

      (2) Developing plans as the project progresses to the implementation phase;

      (3) Managing Risk Assessment (AFT0170);

      (4) Reviewing contract specifications (AFT0430); and

      (5) Producing timelines for project management and plan.

   c. (Med) Managing project implementation to include:

      (1) Initiating project implementation;

      (2) Performing phase reviews to include:

(a) Monitoring contract progress against milestones (AFT0270);

(b) Managing short work periods (AFT0210);

(c) Validating contract refit specifications and drawings (AFT0670); and

(d) Managing contract compliance (AFT0010).

d. (Med) Managing the project closeout to include:

(1) Performing project closures; and

(2) Drafting a project AAR.

e. (Med) Reporting on the Engineering Change (EC).

4. References. To be determined by civilian instructor.

5. Task Numbers. AFT0010; AFT0130; AFT0160; AFT0170; AFT0210; AFT0270; AFT0430; AFT0670; AFT0680; AFT1110; AFT1410.

6. Training Limitations.

a. PO 005 to be instructed by a project management SME (civilian instructor or military member with project management training and experience) with support from a MAR TECH PO1 (or above) / HOD qualified MSEO for technical knowledge.

b. PO 005 will have an overall PC based on an evaluation of a current or in progress Engineering Change (EC) that the students will gather information prior to and during the NCR visit. This NCR visit will be as laid out in EO 005.04.

PO 005 ***Manage DND Projects***

EO 005.01 **Initiate a Project Proposal**

1. Performance. Initiate a Project Proposal.

2. Conditions.

   a. Given:

      (1) DWAN Workstation;

      (2) Message Drafter;

      (3) Message template (MARPACORD 2-13); and

      (4) References.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall initiate a project proposal IAW refs by:

   a. (Low) Identifying project terminology.

   b. (Med) Completing a feasibility study by:

      (1) Identifying conflicting timelines;

      (2) Identifying alternative solutions;

      (3) Identifying a minimum of 2 COAs;

      (4) Recommending a preferred solution using SMART (Specific, Measurable, Attainable, Realistic, and Time bound) criteria; and

      (5) Managing risk analysis (AFT0160).

   c. (Med) Identifying a project charter.

   d. (Med) Establishing TORs for project team members.

   e. (High) Identifying required project facilities. and

   f. (High) Establishing project timelines by:

      (1) Defining project timeline limitations; and

      (2) Defining project phases.

4. Time Required.

   a. EO – 325 mins to deliver training; and

   b. EC – 45 mins (total for all trainees).

5. Remarks.

   a. PO 005 to be instructed by a project management SME (civilian instructor or military member with Project management training and experience) with support from a MAR TECH PO1 (or above) / HOD qualified MSEO for technical knowledge.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Identify project terminology			
1. Define: a. Project definition. b. Project management definition. c. Project life cycle (DND phases): i. Initiation; ii. Implementation; and iii. Closeout. d. Gantt chart. e. Scope. f. Stakeholder. g. Critical path. h. Milestones. i. Feasibility study. j. TOR. k. Charter. l. Business case.		**Method / Media**: handout, F2F (Interpreted Lecture) **Trg Environment**: classroom **Activity Details**: Trainees will be provided a handout with a list of definitions. Trainees will be called upon during class to re-define the terms in their own words. **Trg Resources:** • Handout • PMP references	50
LESSON OBJECTIVE: Complete feasibility study			
1. Identify conflicting timelines. 2. Identify alternative solutions available. 3. Identify different COAs: a. Benefits. b. Costs. c. Manning. d. Resources. 4. Recommend preferred solution. 5. Manage risk analysis (AFT0160): a. Personnel safety. b. Materiel – this is normally done for the individual equipment/system level and at the platform level. c. Mission. d. Environmental protection. 6. Develop SMART criteria: a. Specific. b. Measurable. c. Attainable. d. Realistic. e. Time bound.	NAVORD 3001-1	**Method / Media**: EDI **Trg Environment**: Classroom **Activity Details**: Instructor will explain and demonstrate the procedure for performing a feasibility study and risk analysis IAW the Operational Benefit Decision Matrix (OBDM) IAW ref. Trainees will create a feasibility study using a supplied simple scenario that will include at least two COAs. Trainees will be paired up and exchange studies for peer feedback. **Trg Resources:** • Workstation (DWAN) • Study and OBDM Templates • Scenario	30 (ED) 100 (I + feedback)

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
7. Documenting results in a feasibility report.			
LESSON OBJECTIVE: Identify project charter			
1. identifying the project vision and objectives. 2. defining the complete scope of the project. 3. listing all the critical project deliverables. 4. stating the customers and project stakeholders. 5. listing the key role and the responsibilities. 6. creating an organizational structure for the project. 7. documenting the overall implementation plan; and 8. listing any risk, issues, and assumption.		**Method / Media**: F2F **Trg Environment**: Classroom **Activity Details**: Instructor to provide brief synopsis of teaching points and how they relate to military projects. Provide a copy of an existing charter (eg. FELEX). **Trg Resources:** Example charter	30
LESSON OBJECTIVE: Establish Terms of Reference			
1. Defining the real purpose of each role. 2. Listing the key responsibilities of each role. 3. Defining to whom each role will be reporting. 4. Creating a detailed Organizational chart. 5. Listing the skills and experience needed for each role. 6. Defining any relevant qualifications for each role. 7. Setting out the key performance criteria for each role. 8. Compare civilian to military TORs. 9. Identify Project Staff.		**Method / Media**: F2F **Trg Environment**: Classroom **Activity Details**: Instructor to provide brief synopsis of teaching points and how they relate to military projects. Provide a copy of existing TORs as examples. **Trg Resources:** example TORs	30

LESSON OBJECTIVE: Identify project facilities			
1. Coordinate facility requirements for contractors (AFT1110). 2. Ensure correct infrastructure is available.		**Method / Media**: GD **Trg Environment**: Classroom **Activity Details**: Instructor will guide trainees through a discussion on the teaching points and how they relate to military projects. The instructor will provide a copy of existing PMO facility checklist templates/examples, and ensure the discussion covers off all TPts. **Trg Resources:** PMO facility checklist templates/examples	30
LESSON OBJECTIVE: Identify project timeline			
1. Define project timeline limitations. 2. Identify financial constraints. 3. Identify deliverable timelines. 4. Identify risks and impact to the project. 5. Identify scope-creep. 6. Define timeline feasibility. 7. Define project phases. 8. Draft ship's SWP response message (from template).	MARPA CORD 2-13	**Method / Media**: GD, templates **Trg Environment**: Classroom **Activity Details**: Instructor will guide trainees through a discussion on the teaching points and how they relate to military projects. The instructor will provide copies of existing timelines with various restrictions as examples, and ensure the discussion covers off all TPts. MAR TECH Instructor will show an example ship's SWP response message and walk trainees through it. **Trg Resources:** • Examples of PM timelines • Ship's SWP response message template	40 (GD) 15 (Template)

7. Test Details. EO 005.01 will be tested in EC 005.01 IAW table below and Annex K.

EC 005.01: Initiate a Project Proposal

EC Time: 40 mins + 5 mins debrief

ASSESSMENT ACTIVITY: Trainee will be given the scenario's SWP Alert Message (T -60) and ship's schedule for a short work period (ship will be available). Scenario will include a replacement of a major piece of equipment (GT, pump, rod, etc.). Trainee will have to draft the response (ship's SWP response message / T -45) for the MSE section.
Pass/fail criteria: 100% on checklist
Scenario to include:

- Departmental OPI;
- Work requests;
- Conflicting activity;
- Planned trials;
- Ships movement;
- Fueling/defueling/ammo/de-ammo; and
- Misc info (for message).

ASSESSMENT RESOURCES:

- Scenario;
- Workstation (DWAN);
- Message template (MARPACORD 2-13);
- Message Drafter; and
- Printer.

IMPACT OF FAILURE: Remedial Training / Re-test.

MAR TECH RQ-PO1 – EC 005.01 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all checklist items to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial Training / Re-test

ITEM	SAT	UNSAT	COMMENTS
1. Draft message			
OPI and contact # for each department			
Work carried out by SS			
Operational, ceremonial, or social events that may conflict			
Planned basin/equipment trials			
Scheduled ships movements			
Special directions			
Miscellaneous information (must be considered)			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**
THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 005 Manage DND Projects

EO 005.02 Create a Project Plan

1. Performance. Create a Project Plan.

2. Conditions.

   a. Given:

      (1) DWAN workstation;

      (2) MS Project;

      (3) Examples and templates;

      (4) Operational Benefit Decision Matrix (risk assessment chart); and

      (5) References.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall create a project plan IAW refs by:

   a. (Low) Utilizing project management software if applicable.

   b. (Med) Developing a resource plan to include validating equipment specifications (AFT0680).

   c. (Low) Developing a financial plan.

   d. (High) Managing risk assessment (AFT 0170) to include:

      (1) Establishing the nature of the risk from the risk analysis;

      (2) Determining the likelihood of the risk occurring;

      (3) Determining the impact of the risk; and

      (4) Generating an Operational Benefit Decision Matrix IAW NAVORD 3001-1.

   e. (Low) Developing a communication plan.

   f. (Med) Developing a procurement plan.

   g. (Med) Developing an acceptance plan.

   h. (High) Reviewing contract specifications (AFT0430).

   i. (High) Producing timelines for project management by:

      (1) Describing Gantt chart components;

      (2) Identifying scope and scope creep; and

      (3) Populating a Gantt chart.

4. Time Required.

   a. EO – 805 mins to deliver training; and

b. EC – 570 mins (total for all trainees).

5. Remarks.

a. PO 005 to be instructed by a project management SME (civilian instructor or military member with Project management training and experience) with support from a MAR TECH PO1 (or above) / HOD qualified MSEO for technical knowledge.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Utilize Project management software			
1. MS Project. 2. DRMIS Project builder. 3. MS Excel.		**Method / Media**: EDI **Trg Environment**: Classroom **Activity Details**: Instructor will ED different software suites used to build project manage components. Trainees will follow along and practiced based on given data. **Trg Resources:** • Workstation (DWAN) with DRMIS • MS Project • Instructor qualified in MS Project preferred	140
LESSON OBJECTIVE: Develop a resource plan			
1. Identify resources to include: a. Types of labour required for the project. b. Personnel required to fill each role. c. Equipment needed. d. Material requirements. 2. Validate equipment specs (AFT0680).		**Method / Media**: F2F, GD **Trg Environment**: Classroom **Activity Details**: Instructor to provide examples of resource plans. Trainees to discuss providing resources based on prior experience, instructor to ensure all teaching points covered. **Trg Resources:** examples of resource plans	40

LESSON OBJECTIVE: Develop a financial plan			
1. Establish type of labour costs to be incurred during the project. 2. Identify equipment needed to deliver the project: a. Procurement; and/or b. Leasing. 3. List various materials needed by the project. 4. Establish unit costs for labour, equipment, and materials. 5. Establish other cost types such as administration. 6. Identify amount of contingency needed (usually 10%).		**Method / Media**: F2F, GD **Trg Environment**: Classroom **Activity Details**: Instructor to provide examples of financial plans. Trainees to discuss providing resources based on prior experience, instructor to ensure all teaching points covered. **Trg Resources:** examples of financial plans	40
LESSON OBJECTIVE: Manage risk assessment (AFT0170)			
1. Establish nature of the risk from the risk analysis: a. Personnel safety. b. Materiel – this is normally done for the individual equipment/system level and at the platform level. c. Mission. d. Environmental protection. 2. Determine likelihood of the risk occurring. 3. Determine impact of risk. 4. Generate Operational Benefit Decision Matrix (risk assessment chart)	NAVORD 3001-1 In-Service Naval Materiel Risk Management - Process	**Method / Media**: F2F, GD **Trg Environment**: Classroom **Activity Details**: Instructor to provide examples of risk assessment matrix. Trainees to discuss providing resources based on prior experience, instructor to ensure all teaching points covered. Trainees to create an Operational Benefit Decision Matrix to be used for EC and PC. **Trg Resources:** • examples of Operational Benefit Decision Matrix • references • workstation (DWAN)	100
LESSON OBJECTIVE: Develop communication plan			
1. Identify stakeholder communications need. 2. Schedule periodic progress update meetings.		**Method / Media**: Brainstorm **Trg Environment**: classroom **Activity Details**: Trainees to brainstorm methods of communication with stakeholders IAW teaching points. **Trg Resources:** Whiteboard/flipchart	10
LESSON OBJECTIVE: Develop procurement plan			
1. Define project procurement requirements: a. Material. b. Equipment.		**Method / Media**: Brainstorm **Trg Environment**: classroom	20

c. Infrastructure. 2. Identify suppliers. 3. Identify delivery details.		**Activity Details**: Trainees to brainstorm procurement IAW teaching points. **Trg Resources:** Whiteboard/flipchart	
LESSON OBJECTIVE: Develop acceptance plan			
1. Establish HAT/SAT timelines. 2. Identify approving authorities. 3. Identify documents for acceptance. 4. Identify process for non-acceptance.		**Method / Media**: F2F **Trg Environment**: Classroom **Activity Details**: Instructor to provide examples of acceptance form processes. **Trg Resources:** • Example of Certificate of Compliance • Example of HAT	15
LESSON OBJECTIVE: Review Contract specifications (AFT0430)			
1. Identify a statement of requirement (SOR). 2. Review statement of work (SOW). 3. Review deliverables provided by the supplier. 4. Identify training, documentation and support provided by the supplier. 5. Review contractual terms and conditions.		**Method / Media**: F2F **Trg Environment**: Classroom **Activity Details**: Instructor to provide examples of SOR, SOW, and contract specs and discuss. **Trg Resources:** Examples of SOR, SOW, contract specs.	40
LESSON OBJECTIVE: Produce timelines for project management			
1. Describe Gantt chart components: a. Dates b. Tasks c. Timeframes d. Milestones e. Dependencies (Critical Path) f. Progress g. Current date h. Task identifiers i. Resources j. Slack-time 2. Identify scope and scope creep: a. Arising's (Growth work) b. Material delays c. Project redirection/vision change 3. Populate Gantt chart.		**Method / Media**: EDIP **Trg Environment**: Classroom **Activity Details**: Instructor to ED TPts. Trainees to populate a Gantt chart based on supplied data. Instructor to provide feedback. **Trg Resources:** • Gantt chart example • Workstation (DWAN) • MS Project • Scenario tasks (simple)	400

7. Test Details. EO 005.02 will be tested in EC 005.02 IAW table below and Annex K.

EC 005.02: Create a Project Plan

EC Time: 400 mins + 12 mins brief per trainee + 5 min debrief per trainee (Total=570 mins)

ASSESSMENT ACTIVITY: In groups, Trainees will draft a Gantt chart PMP based on the SWP proposal from EC 005.01 and will provide a project brief.
Pass/fail criteria: 70% correct

ASSESSMENT RESOURCES:
- PMP brief template; and
- MS Project.

IMPACT OF FAILURE: Remedial Training.

MAR TECH RQ-PO1 – EO 005.02 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a 70% SAT on all checklist items
2. IMPACT OF FAILURE: Remedial Training / Re-test

ITEM	SAT	UNSAT	COMMENTS
1. Create project timeline briefing			
Stakeholders listed			
Deliverables listed			
Contractors listed			
Risks listed			
Critical path identified			
Milestones identified			
Slack-time identified			
Phases defined			
Resources identified			
Tasks defined by phase			
Start and end states defined			
Gantt chart completed			
2. Present project plan to stakeholder			
Clarity			
Concise			
Accuracy			
Ability to answer questions			
Maintain time constraints (10 min +/- 2min)			
Matches PMP template			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 005 ***Manage DND Projects***

EO 005.03 Manage Project Implementation

1. Performance. Manage project implementation.
2. Conditions.
   a. Given:
      (1) Gantt chart;
      (2) DWAN Workstation;
      (3) MS project;
      (4) Scenario; and
      (5) References;
   b. Denied: Nil;
   c. Environmental: Classroom.
3. Standard. The trainee shall manage project implementation IAW refs by:
   a. (Med) Initiating project implementation to include:
      (1) Establishing project start conditions;
      (2) Reaffirming stakeholder readiness; and
      (3) Manage a kick off meeting with contractors and stakeholders.
   b. (Med) Reviewing phases to include:
      (1) Monitoring contract progress versus milestones (AFT0270);
      (2) Managing Short Work Periods (AFT0210) when applicable;
      (3) Validating contract refit specifications and drawings (AFT0670); and
      (4) Managing contract compliance (AFT0010).
4. Time Required.
   a. EO – 250 mins to deliver training; and
   b. EC – 220 mins (total for all trainees).
5. Remarks.
   a. PO 005 to be instructed by a project management SME (civilian instructor or military member with Project management training and experience) with support from a MAR TECH PO1 (or above) / HOD qualified MSEO for technical knowledge. Civilian style examples are used, and military support team and trainees discuss specific military context implications.
6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Initiate project implementation			
1. Establish start conditions. 2. Reaffirm stakeholder readiness. 3. Manage a kick off meeting with contractors and stakeholders to: a. Establish a process for recording time within projects. b. Identify and resolve time management issues. c. Managing project timeline (AFT0130). d. Identify quality assurance measures: i. stop checks; and ii. ITPs. e. Identify requests for change. f. Confirm the feasibility of each change.		**Method / Media**: Role Play **Trg Environment**: Classroom **Activity Details**: Trainees will break into two teams. Team members will randomly draw one role each (contractor, supplier, stakeholder, project manager, etc.) with designated "issues", equipment, resource requirements, etc. They will act out a kick off meeting to present their specific issues. The designated project manager will have to manage the various inputs from the meeting. Instruct to monitor, provide feedback, and inject random issues. **Trg Resources:** • scenario including role cards/specifications • large format Gantt chart	200
LESSON OBJECTIVE: Perform phase reviews			
1. Manage progress meetings by: a. Monitoring contract progress verse milestones (AFT0270). b. Reviewing timesheets/DRMIS to monitor the time spent by staff. c. Manage Short Work Periods (AFT0210). 2. Manage deck-plate meetings: a. Coordinate ship staff participation. b. Identify issues raised during progress meeting (timeline). 3. Manage Stop checks by: a. Undertake quality control. b. Validate contract refit specification and drawings (AFT0670). c. Manage contract compliance (AFT0010).		**Method / Media**: Fishbowl **Trg Environment**: Classroom **Activity Details**: Groups from previous lesson (kick-off meeting) will conduct a progress or deck-plate meeting. Roles will switch within syndicates to allow more trainees to act as project manager. One group will act out a progress meeting and the other will act out a deck-plate meeting in front of the other group. During the meetings, they will have to pinpoint the stop-checks (IAW TPts). On completion the non-acting group will debrief the acting group on how they felt the meeting went, focusing on the lesson TPts. **Trg Resources:** • scenario including role cards/specifications • large format Gantt chart	20/group 5 debrief/ group (50 total)

7. Test Details. EO 005.03 will be tested in EC 005.03 IAW table below and Annex K.

EC 005.03: Manage project implementation

EC Time: 100 mins (concurrent) + 10 mins back-brief per trainee

ASSESSMENT ACTIVITY: Trainees to modify timelines in EC 005.02 due to changing conditions provided by instructor. Trainees will each be given 3 random changes (pulled out of a hat) they will have to mitigate. Trainees will back brief the class on their mitigations to stay within project limitations. Instructor may ask clarification questions during back brief.
Pass/fail criteria: SAT on all Critical checklist items
Changing conditions could include:

- Change of command / Dress-ship / ceremonial;
- Ammo/de-ammo / Fuel/de-fuel;
- Shift ship;
- Base-wide exercise;
- Material delays / equipment problems;
- Reallocation of RF priorities (submarine); and
- RadHaz / Divers.

ASSESSMENT RESOURCES:

- Scenario modifications;
- Workstation (DWAN access);
- Notes from EC 005.02;
- References; and
- MS Project.

IMPACT OF FAILURE: Remedial Training.

MAR TECH RQ-PO1 – EO 005.03 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a SAT on all Critical checklist items (marked with an asterisk [*]).
2. IMPACT OF FAILURE: Remedial Training / Re-test

ITEM	SAT	UNSAT	COMMENTS
3. Modify project timeline			
Stakeholders listed			
Deliverables listed			
Contractors listed			
Risks listed			
Critical path identified			
Milestones identified			
Slack-time identified			
Phases defined			
Resources identified			
Tasks defined by phase			
Start and end states defined			
* Gantt chart modified but still satisfies original end date			
* Justification of modifications			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 005 ***Manage DND Projects***

EO 005.04 **Manage project closeout**

1. Performance. Manage project closeout.
2. Conditions.
   a. Given:
      (1) AAR template;
      (2) Scenario;
      (3) DWAN workstation;
      (4) MS Project; and
      (5) References;
   b. Denied: Nil.
   c. Environmental: Classroom.
3. Standard. The trainee shall manage project closeout IAW refs by:
   a. (Med) Performing project closures to include:
      (1) Identifying project completion criteria;
      (2) Listing any outstanding activities or deliverables;
      (3) Creating a plan for passing deliverables to customer;
      (4) Planning handover of project documentation;
      (5) Closing supplier contracts and agreements;
      (6) Releasing project asset and materials to CFSS/Disposal; and
      (7) Communicating closure of the project.
   b. (Med) Drafting a project AAR to include:
      (1) Measuring the benefits and objectives;
      (2) Deciding whether the project was within scope;
      (3) Evaluating contractor process (AFT1410);
      (4) Assessing the final deliverables produced;
      (5) Reviewing the project against schedule;
      (6) Comparing the expenditure against budget;
      (7) Stating the final outcome of the project;
      (8) Identifying the key project achievements and milestones; and
      (9) Documenting lessons learned for future projects.
4. Time Required.
   a. EO – 100 mins to deliver training, 2000 mins NCR tour;

b. EC – 440 mins (total for all trainees); and

c. PC – 400 minutes (for four groups).

5. Remarks.

a. NCR Tour is to be arranged the first week of the course by the course staff. All trainees will need to provide their TIN for bookings.

b. Trainees will be divided into two groups and will identify a current or in progress Engineering Change at the start of the course. During the NCR visit the groups will have an opportunity to liaise with the applicable SMEs (ie LCMM). The two groups will each submit a memorandum detailing where the identified EC is in the process and issues related to implementation as detailed in the PC checklist. The two groups will each provide a 30-minute presentation detailing their findings.

c. PO 005 to be instructed by a project management SME (civilian instructor or military member with Project management training and experience) with support from a MAR TECH PO1 (or above) / HOD qualified MSEO for technical knowledge.

d. Board recommends that EC 005.04 is completed after first-hand exposure to the project management process during the NCR tour.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Perform project closures			
1. Identify project completion criteria. 2. List any outstanding activities or deliverables. 3. Create a plan for passing deliverables to customer. 4. Plan handover of project documentation. 5. Close supplier contracts and agreements. 6. Release project asset and materials to CFSS/Disposal. 7. Communicate closure of the project.		**Method / Media**: GD **Trg Environment**: Classroom **Activity Details**: Trainees to discuss closing out a project under the direction of the instructor, hitting all teaching points. **Trg Resources**: Nil	50
LESSON OBJECTIVE: Draft project after action report			

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
1. Decide whether the project was within scope. 2. Evaluate contractor process (AFT1410). 3. Assess the final deliverables produced. 4. Review the project against schedule. 5. Compare the expenditure against budget. 6. State the final outcome of the project. 7. Identify the key project achievement and milestones. 8. Document any lessons learned for future projects. 9. Complete RF post project survey. 10. Communicate project's success to stakeholders.		**Method / Media**: GD **Trg Environment**: Classroom **Activity Details**: Trainees to discuss a project AAR under the direction of the instructor, hitting all teaching points. Instructor to show an example AAR and FMF-0588 (Ships Customer Satisfaction Survey) **Trg Resources:** • FMF-0588 (Ships Customer Satisfaction Survey) • AAR template for work period	50
LESSON OBJECTIVE: Describe the capabilities of DGMEPM and NETE			
1. Describe key positions within: a. Director General Naval Strategic Readiness (DGNSR), to include: i. Director Naval Personnel and Training (D Nav P &T 3). b. Director General Naval Force Development (DGNFD), to include: i. Director Naval Requirements (DNR); ii. Director Naval Major Crown Projects (DNMCP); and iii. Director New Capability Introduction (DNCI). c. Director General Maritime Equipment Management (DGMEPM), to include: i. Chief of Staff Maritime Equipment Program Management (COS MEPM);		**Method / Media**: Tour/SME presentations **Trg Environment**: Various offices (TBD while scheduling) **Activity Details**: Trainees will travel to NCR and visit with key personnel in DGMEPM and NETE organizations. They will be exposed to the upper-level management positions of DGNSR and DGNFD, highlighting the integration of public works and DND as well as the systems approach to procurement/project management processes. They will discuss with project managers to see first-hand how to manage higher level projects. They will strengthen their support network prior to having the responsibility of their own department. Instructor to liaise ahead of time with POCs to schedule meeting with guest speakers. **Trg Resources:**	2000 (includes tours and transport to NETE) Transport to NCR Sun / Transport from Montreal (NETE) Sat

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
ii. Director Maritime Management and Support (DMMS); iii. Canadian Forces Maritime Experimental and Test Ranges (CFMETR); iv. Naval Engineering Test Establishment (NETE); v. Director Naval Platform Systems (DNPS 2,3,6, 7 &8); vi. Director Maritime Equipment Program Management (Major Surface Combatant) (DMEPM(MSC)); vii. Director Maritime Equipment Program Management (Submarines) (DMEPM(SM)); and viii. Director Maritime Equipment Program Management (Minor War Vessels & Auxiliaries) (DMEPM(MWVA)). 2. Describe key position responsibilities and levels of authority. 3. Develop networking skills. 4. Detail the systems approach to procurement and project management processes.		• POC (Cox'n DGMEPM) • SME presentations	

7. Test Details. EO 005.04 will be tested in EC 005.04 IAW table below and Annex K.

EC 005.04: Manage project closeout

EC Time: 200 mins draft AAR / prepare brief (concurrent); 15 mins brief + 5 mins debrief per trainee

ASSESSMENT ACTIVITY: Trainees will draft a project AAR and provide a project closeout brief to the class based on their products of EC 005.01, 005.02, and 005.03.
Pass/fail criteria: 60% SATs to be successful
ASSESSMENT RESOURCES:

- Scenario
- Notes from EC 005.01, 005.02, and 005.03, NCR tour
- References
- DWAN workstation
- MS Project

IMPACT OF FAILURE: Remedial Training / Re-test

MAR TECH RQ-PO1 – EC 005.04 Assessment Checklist			
Candidate Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Trainees must achieve a total of 60% SATs to be successful with this assessment.
2. IMPACT OF FAILURE: Remedial Training / Re-test

ITEM	SAT	UNSAT	COMMENTS
1. AAR			
Identified if project fell within scope			
Evaluated contractor process			
Assessed final deliverables			
Compared project timeline against original schedule			
Compared the expenditure against budget as required			
Stated the final outcome of the project (eg. complete, incomplete, cancelled)			
Identified the key project achievements			
Identified the key project milestones			
Identified lessons learned for future projects			
2. Brief			
Supporting visual aids enhanced the presentation (if used)			
Time was 15mins (+/- 3mins)			
Information was presented in a logical order			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 005: Report on EC/UCR Project based on NCR Visit

PC Time: 200 mins prepare brief based on NCR visit and group memo; 30 mins brief + 10 min questions +10 mins debrief per group

ASSESSMENT ACTIVITY: Trainees will be divided into groups of 3-4 to report on a chosen and approved EC/UCR and provide a project closeout brief to the class based on the NCR visit.

Pass/fail criteria: 60% SATs to be successful

ASSESSMENT RESOURCES:

- Approved EC/UCR project;
- Notes from NCR tour;
- References;
- DWAN workstation; and
- MS Project.

IMPACT OF FAILURE: Remedial Training / Re-test

MAR TECH RQ-PO1 – PC 005 Assessment Checklist			
Group Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:
1. PASS / FAIL CRITERIA: Trainees must achieve a 70% SAT on all checklist items
2. IMPACT OF FAILURE: Remedial Training / Re-test

ITEM	SAT	UNSAT	COMMENTS
4. Create project proposal (EC/UCR)			
Feasibility Study (Who? How?)			
Project Charter (Purpose of EC/UCR)			
Terms of Reference (Stakeholders identified)			
Project Facilities (Units Involved)			
Project Timeline (Gantt Chart)			
Risks identified/feasibility of implementation			
5. Project plan			
Overview			
Identify Software Used			
Identify management of resources (who is supplying/installing)			
Financial Plan (Estimated Cost)			
Risk Assessment (Common EC/UCR risks)			
Comms Plan (Lines of communication)			
Procurement Plan (Who is purchasing/what process)			
Acceptance Plan (STW/Tests/Trials)			
Contract Spec (Life Cycle requirements for this EC)			
6. Present project plan to stakeholder (assessor) (Through memo and group presentation)			
Clarity			
Concise			
Accuracy			
Ability to answer questions			
Maintain time constraints (30 min +/- 5min)			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE CANDIDATE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 006 Advise Command

1. Performance. Advise Command

2. Conditions

   a. Given:

      (1) References;

      (2) Assistance;

      (3) Operational ship; and

      (4) Computer station with DWAN access.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The MAR TECH manager shall independently advise command on:

   a. Machinery and System Failures by:

      1) Identifying defect;

      2) Determining cause of failure;

      3) Explaining current COA with unresolved defect;

      4) Explaining current effect on system and mission;

      5) Developing mitigation strategies;

      6) Proposing COAs for resolution with RA;

      7) Utilizing supporting Agencies;

      8) Utilizing onboard liaison; and

      9) Reporting estimated time of repair.

   b. Emergent machinery/equipment failures during DC situations, and Action Stations by:

      (1) Determining the situation and failure;

      (2) Assessing the impact of the failure based on the risk matrix;

      (3) Determining options;

      (4) Recommending solution; and

      (5) Reporting estimated time of repair.

4. References. A7, A8, A9, A10, A13, A14, A18, and A15.

5. Task, Skill, Knowledge Numbers. AFT0100, AFT0170, AFT0090, AKT1240, ZT0070, AET1010, and AFT0210.

6. Training Limitations. Scheduling: The timing of this PO should occur prior to the planned field trip to the NCR to allow the trainees to capitalize on the material contained within.

EO 006.01 Advise on Machinery and System Failures

1. Performance. Advise on Machinery and System Failures

2. Conditions

   a. Given:

      (1) References;

      (2) Assistance; and

      (3) Computer station with DWAN access.

   b. Denied: Nil

   c. Environmental: Classroom

3. Standard. The trainee shall independently advise on machinery and system failures IAW refs by:

   a. Identifying defect.

   b. Determining cause of failure.

   c. Explaining current COA with unresolved defect.

   d. Explaining current effect on system and mission.

   e. Developing mitigation strategies.

   f. Proposing COAs for resolution with RA.

   g. Utilizing supporting Agencies.

   h. Utilizing onboard liaison.

   i. Reporting estimated time of repair.

4. Time Required.

5. Remarks. Nil.

6. Teaching Points.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
LESSON OBJECTIVE: Advise on Machinery Equipment Failures			
1. Simplified way of presenting problem: a. Use of drawing. b. Location in system. c. Design function. d. Degraded/loss of function. e. Result/impact of fault. f. Current operational impact(s).	A7, A8, A9, A10, A13, A14, A18, A15.	**Method / Media:** IL **Trg Environment:** Classroom **Activity Details:** • Instructor will present a defect to the trainees in the desired format. It will be a slow time delivery with enhancement on the TPs. The format will be presented on a white board or other suitable	700 mins + 360 mins for embedded ECs.

Teaching Points	Ref	Method / Media / Environment / Training Activity Details	Time (mins)
2. Failure cause: a. Known or unknown. b. Current troubleshooting efforts. c. Independent or resultant failure. 3. Resolution development: a. COAs available. b. Pros and cons of each COA. c. Mission, personnel, safety, environmental, equipment effects. d. RA chart. e. Advisory agencies/pers: i. FER; ii. FTO/FTC; iii. FTA; iv. FMF Engineering, Elect, NAO; v. FMF Production; vi. PL/SSO; vii. DA, LCMMs, DNPS; viii. Blog via Log dept; and ix. FSE. f. Internal stakeholders: i. CHODs, Coxn, XO, CBTO, Flex considerations, NAVO, LOGO.		medium. The presentation will show a diagram which explains the defect, the impact on the current programme, recommended COAs with pros and cons, a RA chart for assessing each COA, a recommended COA, as well as outside agencies that could or have been engaged in resolution or mitigation of this situation. For the EC, a defect will be selected based on the trainee's technical background, ie: Mech or Elect. The students will then present their equipment failure to the class in an interactive forum that re-enforces the TPs.	

EO 006.02 Advise on Emergent Machinery/Equipment Failures

1. Performance. Advise on Emergent Machinery/Equipment Failures

2. Conditions

   a. Given:

      (1) References;

      (2) Assistance; and

      (3) Computer station with DWAN access.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall independently advise on emergent machinery/equipment failures during DC situations, and Action Stations IAW refs by:

   a. Responding to alarms.

   b. Analysing DC effects on propulsion, ancillary, auxiliary, and electrical systems to include:

      (1) Identifying defect;

      (2) Determining cause of failure;

      (3) Explaining current COA with unresolved defect; and

      (4) Explaining current effect of system and mission.

   c. Overseeing safety and environmental considerations such as during a spill response.

   d. Developing mitigation strategies.

   e. Proposing COAs for resolution.

   f. Utilizing supporting agencies.

   g. Utilizing onboard liaison.

   h. Reporting estimated time of repair.

4. Time Required.

   a. EO –200 mins to deliver training; and

   b. EC – 100 mins.

5. Remarks. Nil.

6. Teaching Points.

LESSON OBJECTIVE: Advise on Emergent Machinery/Equipment Failures			
1. DC Picture Interaction:	A7, A8,	**Method / Media: GS, IL**	

a. Liaison with HQ1 on DC actions including: i. Machinery space considerations; ii. Use of MCR manning/resources; iii. Effect of power isolation/electrical damage on engineering systems; iv. Casualty power requirements wrt safety, routing, restoration priority; v. Effect of damage to mechanical engineering systems; vi. Environmental and safety aspects. 2. During discussion and scenario, consider the following: a. Responding to alarms; b. Analysing DC effects on propulsion, ancillary, auxiliary, and electrical systems; c. Providing oversight on environmental considerations such as during a spill response; d. Providing oversight on safety considerations; e. Developing mitigation strategies; f. Proposing COAs for resolution; g. Utilizing supporting Agencies; h. Utilizing onboard liaison; and i. Reporting estimated time of repair.	A9, A10, A13, A14, A18, A15.	**Trg Environment:** Classroom/IPMS Trainer **Activity Details**: Interactive Lecture • Activity is designed to develop the role of the MSEC during action/emergency stations and/or engineering drills where the maintenance of propulsion is vital. The activity will focus on the role of SME co-ordinator in the MCR. Trainees will be shown engineering plant drill scenarios that necessitate a quick plan to restore or maintain propulsion/power, action or emergency station scenarios that necessitates co-ordination with the MCR, and combinations of the two. A typical scenario will be demonstrated to the class by first reading the situation aloud, then describing general initial actions and concerns, expectations of senior Mar Tech positions, scenario specific concerns at the CHOD level. This information will be presented visually in a standard format to facilitate discussion and retention. EC will be a written test which examines the trainee's ability to understand his/her role as the MSEC overseeing the MCR picture. For legacy trades, ERs can use any engineering emergency scenario that puts either the maintenance or restoration of propulsion at a premium. For ETs restoration of power scenarios can be used, for HTs, scenarios involving stability or structural damage and the immediate need to stay underway or on course can be used.	Guest lecturers, ST(A), ST(P), CEng, CET. 200 mins Scenario work 100 mins

ANNEX A - *MAIN REFERENCES*

A = CF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other

Ref Code	Publication
A1	Manual of Trim and Stability (all platforms)
A2	Damage Control Manual Vol. 3
A3	D-03-002-008/SG-011 - Requirements for Docking/Undocking of CF Vessels
A4	C-09-153-003/TS-000, Explosive Safety Manual, Volume 3
A5	Naval Engineering Manual – Part 6, Section 4 Docking
A6	Docking plan: HFX-D28-589-000-13 (drawing number)
A7	CFCD 129
A8	NAVORD 1190-2
A9	NAVORD 3000-0
A10	NAVORD 3001-0
A11	NAVORD 3001-1 In-Service Naval Materiel Risk Management - Process
A12	NAVORD 3250-7
A13	CFCD 102
A14	NaMMS Vol. 1 C-03-005-012/AM-001
A15	NMRSS C-23-005-000/AG-001
A16	MARPACORD 2-13
A17	NEM TI-016 (Naval Engineering & Maintenance Technical Instructions) - Materiel Defect Notification: ND Defects for Surface Ships - Process
A18	NEM SOP - FMF Work Period Planning Process (WP3)
A19	Ship's Standing Orders (SSO)
A20	Docking CFTPO
B1	Docking Instructions US Navy S9086-7G-STM-010/CH-997
D1	Docking Master plan (any vessel)
D2	DMCM docking work periods website

ANNEX B - *TASK LIST*

**Highlighted Tasks were determined as "NT" by the JTAR but "T" by the QSPWB IAW decisions in Annex C.*

Task No	Task Statement	Train / No Train	PO
AET1010	Coordinate training and readiness	T	004
AFT0090	Manage maritime evaluation activities	T	004
AFT0010	Manage contract compliance	T	005
AFT0130	Manage project timeline	T	005
AFT0160	Manage risk analysis	T	005
AFT0170	Manage risk assessment	T	005
AFT0210	Manage short work periods	T	005
AFT0270	Monitor contract progress versus milestones	T	005
AFT0430	Review contract specifications	T	005
AFT0490	Review particularized maintenance & repair Schedule (PMRS)	NT	
AFT0530	Review Ship's Standardized Maintenance & Repair Schedule (SSMRS)	NT	
AFT0650	Update the fitted equipment list (FEL)	T	001
AFT0670	Validate contract refit specifications and drawings	T	005
AFT0680	Validate equipment specifications	T	005
AFT0820	Advise on particularized maintenance & repair schedule (PMRS)	T	004
AFT0840	Advise on ship's standard maintenance & repair schedule (SSMRS)	T	004
AFT1010	Complete material certification documentation	NT	
AFT1110	Coordinate facility requirements for contractors	T	005
AFT1410	Evaluate contractor processes	T	005
AFT1520	Investigate major equipment failures	NT	
AKT1190	Manage docking preparations	T	003
AKT1240	Manage operational deficiency report requirement	T	004
CT0090	Coordinate engineering drills	NT	
GT0040	Manage undocking preparations	T	003
IT0120	Assess battle damage to engineering systems	NT	
IT0130	Assess battle damage to vessel	NT	
IT0810	Survey battle damage	NT	
OT0020	Advise on maritime vessel fuelling operations and readiness	NT	
WT0040	Assess ship's stability	T	002

WT0070	Manage ship’s stability	T	002
ZT0070	Manage notice for power	T	004

ANNEX C - *QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS*

QSP Finalization and Review Notes and Recommendations:

For next review/board:
Areas such as stability and DC roles/responsibilities should be coordinated across RQ PO2, USQs, and RQ PO1 so they are tied together, build on previous tasks, and avoid and gaps in training (and tasks, such as when Elec joins at PO1 level, there are gaps in what they have done compared to Mech, but are expected to supervise and perform tasks only the Mech stream have been trained to do).

Dates:

5- 16 Mar 2018

Location:

Haida Bldg. rm 105C, Work Point, CFB Esquimalt

Board Members:

Rank and name	MOSID / Unit	Dates
CPO2 Y. Gagné, Mar Tech TM	00379 / NPTG HQ	Only present for small periods of time throughout board.
CPO2 J. Drew, Chair	00379 / NTDC(P)	5-16 Mar 18
CPO2 S. Wilcox, SME	00379 /ADM Mat/DGPM	5-16 Mar 18
PO1 J. Millie, SME	00379 / NTDC(P)	5-15 Mar 18
PO1 S. Blackbourn, SME	00379 / NTDC(P)	5-15 Mar 18
SLt Adrian Smart, TDO OJT	00211 / NTDC(P)	5-6 Mar 18
SLt Sarah Valentine, TDO	00211 / NTDC(P)	5-16 Mar 18
A/SLt K. Knight, Secretary	00207 / NTDC(P)	7-16 Mar 18

Board Objective:

To analyze the job performance and design the training requirements for MAR TECH Managers based on the JTAR and legacy requirements of the amalgamated technician trades.

Discussions / Decisions:

**Decisions which are stricken through were changed at a later date*

Date	Decision/Event	Detail
5 Mar	Briefing from Command	Cdr Koester (CO NTDC(P)): 1. Key is content so that someone else can build the course 2. Keep doing it well 3. CNPTG coming to brief the team around 1100 today

	Intro brief	Given to SMEs by TDO (chair at appt).
		• Importance of collaboration
		• Mandate: develop training in response to changes to job requirements (amalgamation)
		• Board expectations of QSP:
		  ○ Develop management capabilities in PO1
		  ○ Manager, not supervisor

	CFITES Overview	Given to SMEs by TDO.

	Document review	• Job description for Mech Manager, Elec Manager
		  ○ No USQs at this level
		  ○ Mech and Elec streams melding back together; not the technical expert but will be managing both streams; no new technical training req'd
		  ○ No watchkeeping training (can opt out at PO2 level)
		• RQ PO2 draft QSP PO 002 PO 003
		• JTAR (21 Trained Tasks): Board decided that 3 of the tasks were moved to RQ PO2 level and would be **NT** at RQ PO1 level

	Task analysis: Grouping tasks	
	Brief	To board by TDO
	OT0020	Discussion about whether this is at the PO1 level or whether it's already covered at the PO2 level.
		Board Chair has been part of fuel spills and needed to brief command during these incidents. He would have liked some training on this prior to. He sees the Task as reacting when something goes wrong.
	AFT0650	Discussion about whether this includes "Dismantling". Higher level of DRMIS may be required; may require additional tasks.
		Board reviewed DRMIS QSP: "Dismantling" covered in DRMIS Supervisor Course (Module 2) which is part of RQ-PO2.
		Board decided that "DRMIS Module 3 Maintenance Manager" would be beneficial for the RQ-PO1 course because it covers off some project management required in DRMIS.

Date	Decision/Event	Detail
	General Groups	Project management; DC; DRMIS
	HQ1	~~Board discussed that there is nothing for HQ1 currently; created tasks "Act as DCPO" "Perform as HQ1 team member" "Conduct Power Isolation"~~ **NOTE: HQ1 removed from QSP post-QSPWB*
	Task Analysis: Sequencing tasks	
	Brief	To board by TDO
	Legacy trg	SMEs pulled up QL6 from all three legacy trades to review
		Hull tech: lots of stability which may not be required as Elec.
	PO1 job	Board detailed what the current configuration is for PO1s onboard ship: still one of each legacy trades (2x Mech PO1, 1x Elec PO1); one on each watch plus one "maintenance". There will be no USQ for Elec, but currently (from JTAR) there is a USQ at PO1 level for Mech.
	DC	Board discussed the requirements for the DC portion of the course and whether there would be a need for USQs at this level. Decided to hold off on organizing these tasks until they were able to discuss with TM.
	Project Management	Reviewed the PO from WENG RQ-CPO2 "Manage DND Projects". Looked similar to the old LCC course (including length of 4 days); seemed effective for Mar Tech RQ-PO1 use.
		Board transferred the WENG PO scalar to the wall to see how it matches the tasks we have for the RQ-PO1.
	OT0020 (Advise on fueling)	SMEs decided that it will fall under AFT1520 (technical investigation - TI) in DC phase because they incorporate the same functions (same process, different product)
	CT0090 "Coordinate eng drills"	Discussion surrounding where drills fall in. Training has already been done for creating a training plan etc. at the PO2 level (Coordinate SBT Trg), and this is basically the same only at a higher level. They will already know engineering drills, so no need to train that. Only difference between this and Coordinating SBT trg is that it is at a shipboard level instead of the departmental level, therefore SMEs decided that it is **NT**, experiential.
	Elec/Mech comparison	SMEs went through TOC from legacy QSPs to see if there are any large tasks that are missed in JTAR.
		Hull Tech QL6B: • ~~PO 001 DC: should be reflected in Mar tech RQ-PO1~~ **NOTE: HQ1 removed from QSP post-QSPWB* • PO 002 Stability: should be reflected in Mar tech RQ-PO1

Date	Decision/Event	Detail
		• PO 003 Docking: should be reflected in Mar tech RQ-PO1
		ET QL6B: • ~~Power Isolation~~ **NOTE: Power isolation removed from QSP post-QSPWB*
		Mar Eng QL6: • Docking/Undocking
		Board discussed incorporation of docking/undocking as a PO in RQ-PO1. It is almost guaranteed that you will see a docking at some point on any platform and there's the potential for emergency docking where the PO1 would be the SME.
	Docking/Undocking	Board discussed using the existing PowerPoints to create the new courseware (currently available from PO2 Shane Kendall at NTDC(P) MS Div)
		Board compared the HTech QL6 with the Mar Eng QL7 req'mts
		In JTAR: "Supervise docking/undocking" at PO2 Mech (USQ) level. "Manage" is at the PO1 Mech (USQ) level.
6 Mar	DC	~~Board sees DC running as a week of theory with a week of HQ1 practical.~~ **NOTE: DC removed from QSP post-QSPWB*
	Project management	Board decided project management would be the last PO so that examples from previous POs can be used throughout
	DCPO	~~Board decided on "Act as DCPO". DCPO is the term for the PO1 off-watch (Mech) during emergencies/action stations. The term is from the legacy term "DC CPO".~~ **NOTE: DC removed from QSP post-QSPWB*
		~~Board decided that communications (skill) will be used throughout "act as DCPO" but will be **NT** for experience.~~ **NOTE: HQ1 removed from QSP post-QSPWB*
	Possible PO1 Mech USQ	Board discussed whether Stability and docking/undocking are specific to the PO1 Mech (no need for PO1 Elec to know it). Considered that there is no CPO2 coursing, that Elec doesn't have the basic stability that is taught throughout the Mech stream, that Elec should have the math capabilities (maybe not the actual equations, but the ability to do that level of math)
		TM says that Elec needs to know it as well because when they become the Chief, they'll already have the training.
		LCdr Day (NTDC(P) MS Div Cdr) says that some stability could be moved to the SBT IC trg on RQ-PO2 so that Elec does have a little bit of a background.
		~~Possibly create a PIP / study package for those that are less comfortable with the material so that they can do some studying~~

Date	Decision/Event	Detail
		~~ahead of time~~ [**NOTE: Board decided against a PIP 16 Mar]*. HTech QL5 has an existing PIP for equations which could be used.
		Board (with TM and stakeholder input) decided to include stability and docking/undocking on RQ-PO1; **no PO1 Mech USQ required**.
	Project management	Stakeholder recommends including a contracted-in instructor (out-source)
		TM mentioned tour to NCR to build familiarity with personnel and project management functions at the HQ level (LCMM, Louis-St-Laurent for DGMEPM, Carling Campus for CM, NETE in Montreal). This tour would be roughly 5 training days (flying on weekends) with lectures and tours.
	Docking	Board decided that tasks which were slated as NT on the JTAR (Managing Docking Preparations (AKT1190) and (GT0040) Undocking) are actually to be trained at this level. Originally on the JTAR is was thought it would be trained at the CPO2 level but since there is now no RQ-CPO2, it was moved to the RQ-PO1.
7 Mar	**Design Phase**	
	Design phase brief	Given to board by TDO
	Target audience analysis	Carried out by Board with guidance of TDO. Will be embedded in Annex M of QSP to help with out-source developers understanding of target audience
	Annual throughput / course capacity	Board discussed the projected annual throughput for the course: approximated 10 candidates per course (based on average numbers on legacy training and 10 seats available in BDCS trainer) with 2-3 courses running per year on each coast averaging around 40-50 graduates per year.
	Training Establishment analysis	Carried out by Board with guidance of TDO. Will be embedded in Annex M of QSP to help with out-source developers understanding of TE
	Design: EO Writing	
	EC 002.01 Assess ship's stability	Board decided the pass mark will be 75% because they are given the MT&S and formula sheet and there is a safety aspect.
	CTAT	Any CTAT information or materials are highlighted in Red in Ch. 3. They will need to be developed in house instead of out-sourced. (anything with MT&S including PO 002: Manage Ship's Stability)
	Stability and BDCS	EO 002.02 "Coordinate adjustments to stability": PO1 Kendall was called in as a SME for inputting data into the BDCS as

Date	Decision/Event	Detail
		none of the board members had done it before. TM confirmed the points
	PO Numbering	Board decided that PO 001 would be DRMIS Maintenance Manager course (existing MITE 119098) since it is good to have prior to the other POs
		PO 002 “Manage ship’s stability” because stability will form TPts of two of the follow-on POs, therefore they will need the foundations in this PO prior to PO 003 and PO 004.
		PO 003 “Prepare vessel for docking/undocking”: because it is closely linked to Stability therefore should be taught directly after PO 002.
		~~PO 004 “Manage DC HQ”: because it has a component of stability in it therefore should be taught after PO 002~~ **NOTE: DC removed from QSP post-QSPWB*
		PO 005 “Manage DND Projects”: because it will incorporate scenarios from previous POs therefore should be taught last.
8 Mar	PC 003	The board decided the PC for 003 will be written because dockings are a rare occurrence and would be very costly to simulate the actual event
		The board decided the write up will be similar to the old assessment and the written exam, however no scenario will be added to the written exam because the actions are always the same no matter the situation. They will have to complete relevant portions of a CF 523Certificate of docking and undocking.
		The content of the PC was discussed and it was decided that the 523 would be included for relevant portions to be filled on the exam and references would not be provided. The board believes that at this time it is expected that the candidate knows this information.
		The board discussed the content and decided that the exam would be 30 questions and take 60 min to complete with a 10 min debrief, and will require a passing mark of 70% because it involves a number of safety issues and can cause damage to DND property if done incorrectly. There will be no references provided because the trainee is required to know all this information.
		As a reference the board has included the QL6B Hull Tech PC 002 Exam 1 Docking everything included in the reference shall be represented in the exam that is to be developed.
	EC 003.01	The board discussed different ways to present an EC based on the 523 form and decided that giving them the form is the best way to go about it because there is no problem with testing the 523 in the EC and PC. There was discussion on the passing

Date	Decision/Event	Detail
		mark as well as the number of questions and it was decided that because of the risks to safety and DND property the pass mark was determined to be 80 %. To make the EC useful it was decided that the candidate would be required to reference their answers.
		The board discussed different ways to conduct this exam and decided that a written exam is the best way to present this information.
		The board discussed what information should be on this EC and decided that there was enough information to test the candidate on with minimal redundancy.
	Training Tools (Docking and Undocking)	The board discussed visual aids for this training point and ~~decided that a tank would be best with 3d Printed models IOT properly represent this evolution~~ *[*NOTE: Board decided against a tank was unnecessary 14 Mar]*. It was also discussed that a video or an animated visual aid reference may be a training tool for this Section.
	Training for EO 003.01	The board decided that the first aspect that needs to be covered is the terminology. ~~It was decided that the tank visual aid would be helpful in this teaching point.~~ *[*NOTE: Board decided against a tank was unnecessary 14 Mar]*.
		It was decided that the video in training system "Preparing Ships for Docking" (hull tech. QL 6B) should be included in in this teaching point. Teaching points can be drawn from PO 002.02 of the Hull tech. QSP.
		The board decided that the next teaching point will be "preparation for a docking procedure."
		The board decided to move "docking occasions" to the terminology TP 1
		There was discussion about the name of the EO and it was decided that this lesson was knowledge so the preferred method for delivery is PP stand and deliver. The instructor will just run the candidate's through the points.
	EO 003.02	The board discussed adding a list of requirements to the float procedure however decided that it was unnecessary and the instructor could just go over this information.
	~~PC 004 (DCHQ)~~ **NOTE: HQ1 removed from QSP post-QSPWB*	~~It was decided that TI's and Post event damage will not be part of the PC, because the EC will be sufficient to capture it. The board decided that the passing mark for the PC will be 85% citing safety concerns.~~
		~~The board identified readymade scenarios for the DCPO from the QL6B Syndicate Training guide lines to be used in the development of this section.~~

Date	Decision/Event	Detail
		~~There was discussion amongst the board members as to provide references and resources, it was decided that the candidate may use an "aid memoir" for this evaluation.~~
		~~The board decided that the passing mark of EC 004.0102 DCPO Would be 80% because it is damage control and has little room for error. A critical item checklist is to be drafted and included in the resources for the developers.~~
9 Mar	EC004.02	~~Assessing the electrical manager during emergency stations will be conducted and taught before 004.01 (DCPO) and be tested in the same EO. The decision was made to switch the 2 E0's IOT more effectively test the material.~~ **NOTE: HQ1 removed from QSP post-QSPWB*
	EO 004.01 **NOTE: HQ1 removed from QSP post-QSPWB*	~~The board decided that EO 004.01 will not be assessed on the PC because it is a preparation for experience on ship and an EC is sufficient assessment~~
		~~It was decided that "advise command on post battle damage" has too small of a scope and be added to Submit Post Incident Report 004.03~~
	EC 004.03 (PC 004)	~~Decision by SMES: TI will not be evaluated on the PC and shall only be tested on EC 004.03. (chair later concurred with the decision)~~ **NOTE: HQ1 removed from QSP post-QSPWB*
	FTA	~~The checklist for the EC 004.03 is to be developed from the FTA content that is currently unavailable (Link is broken)~~ **NOTE: HQ1 removed from QSP post-QSPWB*
12 Mar	PC 005 (Project Management)	After some discussion the board decided that the assessment would be based on a "~~short work period~~" *[*NOTE: Was updated to "complete a short work period to replace a piece of equipment"]* because a "docking work period" would be too long and encompass too much material.
		After discussion the board decided that the EC 005.01 assessment would be scenario based on the "ten week letter" and be pass or fail based on a check list.
	NCR Tour	TM has informed the board SMES that their requested NCR tour for the project management section in Ottawa is being reviewed. They are working on the details now and awaiting the approval.
	Capability Investment Development	Link MCDV degaussing (code C.001340) http://cid-bic.forces.mil.ca/CID/Project-Home_e.asp To use during PO 005 for example of the entire brief and the template going forward.
	AFT 0680 (Validate equipment specs)	The board decided that "Validate equipment specs" assumes that it is equipment required for the install vs equipment for replacements. This means that you give the contractor the

Date	Decision/Event	Detail
		proper criteria and ensure that they understand what is required and what the job consists of.
13 Mar	Difference between risk analysis and risk mitigation	The board defined “Risk analysis” as the risks to the project itself. This means risks like time, budget, and labour concerns that can impact the delivery or completion of the project. “Risk assessment” is based on the “risk analysis” planning phase, this is ongoing throughout the project and encompasses all risks that you have the ability to foresee. This is the stage where the project manager mitigates these risks.
	PM Software suite	The board discussed what the various software types are and if the course should include teaching to the full software capabilities (i.e. the MS Project course). The board decided that the course should only teach the introduction and capabilities because going into the details of the different software tools would be far more than the trainee would need at this level.
	Gantt Scenario	After discussion the board decided that the scenario should be simple and should reinforce the process of the Gantt chart. The number of steps is not important so long as they understand what they are doing and why they are doing it
	Gantt Scenario EO 005.03	The board decided that instead of doing a practical EC they are going to modify the results of EC 005.02 by adding changing conditions that will require the trainee to adjust their timeline and Gantt chart IOT simulate real world conditions. The board decided that back briefing the class would be an excellent way for them to show that they can adapt and maintain their project limitations.
	Mitigating Change	After much discussion the board decided that there was no way to effectively teach an individual how to mitigate change, they decided that by the time the trainee has reached this level they should have already gained these skills. After some further discussion the board decided that an effective way to reinforce this experience point would be a role playing activity that would inject problems into the trainee’s plan that require modifications to maintain their schedule.
	Cert 4 consideration	LCdr Day informed the board that this was the end of the road as far as training is concerned and that once the trainees come out of this course they are expected to be able to rank up all the way into chief through experience. There will be no further coursing after this. The board agreed to take another look at the QSP and perhaps insert some more Chief Engineer type activities IOT make sure everything is covered.
	Deck plate meeting	The board defined “Deck plate meeting” to be technical orientated meeting concerning the trade people rather than the

Date	Decision/Event	Detail
		management. “Progress meetings” are more top view that involve the contracts and administration.
	EO 005.03	The board decided to change the EO title “Executing a project plan” to “Manage project implementation” to bring it more in line with project management terminology (the implementation phase).
	EO 005.04	Board decided that ~~no~~ EC was required for this EO because of its small scope. Though it was noted that the scope is more for TPts than EO, it is seen as a separate phase to the project management picture (close out phase) therefore is unique to the other EOs.
	OT0020	Board decided that “Advise on maritime vessel fuelling operations and readiness” was **no train** as it is experiential at the PO2 level.
14 Mar	Stability discussion	Sr WK TNA board joined for a short time to discuss stability throughout MAR TECH training. They agreed at the level of stability training we had laid out for RQ-PO1 and introducing it on the RQ-PO2.
	Review of Mar Eng. QL7	Board reviewed the legacy QSP to confirm that all critical tasks were covered off by the RQ-PO1 since there will not be a RQ-CPO2. Board decided that the only possible gap is with the SLOC (i.e. stores).
	SCA/SLOC	Board decided that SLOC muster would not be trained.
	PM Instructor	The Board decided a PM instructor will be required to teach the entirety of PO 005. This PM instructor can be civilian or military. If military they need to be PM trained through the course and have experience carrying out project management. PM instructor will require a technical advisor for MAR Tech specific questions, this advisor needs to be MAR Tech PO1 or above or HOD qualified Lt(N) MS ENG or above.
	Tank TEL	The Board decided that the stability tank was not a practical idea because it would be too much work and instructors would not look after it. A digital stability modelling software would be the preferred method of instruction because it would demonstrate the process to a higher degree (with more examples) without the maintenance required of a physical model.
	Unscheduled Docking	The board decided it is of high importance to teach the trainee unscheduled docking because in this situations they may not have access to the Dock Master. This means that it is important that the trainee knows how to do it in an emergency situation.
15 Mar	EO 004.03 (Post incident	~~The board decided that they would formulate a checklist for the “draft a TI component” of PO 004.03 to ensure that the trainee covered all the important information. The board decided that~~

Date	Decision/Event	Detail
	reports) **NOTE: DC removed from QSP post-QSPWB*	~~they would provide the candidate with the TOR to complete the form.~~
		~~The board decided that the candidate would be given a scenario to write the EC instead of receiving a BOI report as they originally thought.~~
		~~The board decided that the purpose of the TI exercise was not to have the trainee draft a "perfect" example of a TI, rather the goal is to ensure they are exposed to the process for when they are expected to write one on ship. This is why they made the pass criteria 60%.~~
	Close out Brief (PC 005)	The board decided that there shall be no PC for project management PO because of the large scope and time required to do a project from inception. The board instead decided that the trainee would provide a "project close out brief" to the class in order for the instructor to assess the trainee for EO 005.04
	PC 005.04	The board decided that it is of vital importance that EC 005.04 be completed after the trip to Ottawa because they require exposure to the process of closing a project that can only be effectively gained from Ottawa.
16 Mar	Stability PIP	Board decided there would be no requirement for a PIP / study package for stability because all candidates would have enough math background to be able to successfully complete the stability phase.
	Stability RQ-PO2	Board discussed points from MAR ENG Space 2I/C legacy training to move to the RQ-PO2. Will likely become an EO in PO 002 "Act as SBT I/C".
	WT0040	"Assess Ship's Stability" was found to be a task in the JTAR (matching the main task for EO 002.01). Though not listed at the RQ-PO1 level in the JTAR, it was decided by the board that it would be trained at a different level here to ensure that Elec and Mech managers are at the same level with stability.
END OF QSPWB		
9 May 18	QSP doc alignment post PO 2 Mech/Elec QSPWB	Upon completion of the USQ PO2 Mech/Elec QSPWBs the Elec Supr board determined training on TRP was required. D Nav P&T supported these changes but also indicated that training at the PO1 would have to support CPO2 performance because there is no formal coursing past the RQ PO1. Changes are being made that are being driven from both Mech and Elec separately. To effect these changes to the RQ PO2, the following personnel came together in a working group from 7 – 11 May 18: Lt(N) Robinson, TDO, NTDC(P); Lt Tzotzos, TDO, NTDC(P);

Date	Decision/Event	Detail
		CPO2 Zsombor, Mar Tech Dept Coord; and PO1 Penney, Mar Tech Elec Mgr (NTDC(P). 1. **Mechanical Changes:** Add PO 002 Assess ship's stability back into QSP. 25 April 18, D Nav P&T directed it be removed but as per email below from CPO1 Pryor dated 2 May 18, it was to be put back in. **2. Electrical Changes:** Development of new PO 004 Coordinate Departmental Readiness Requirements. These POs will be added and will include a number of tasks identified by CPO1 Pryor that support CPO2 performance. Email below identifies D Nav P&T requirements: **From:** Pryor CPO 1 BA@C Navy D Nav P&T@Ottawa-Hull **Sent:** May-03-18 8:33 AM **To:** Tzotzos Lt LA@NPTG NTDCP@Esquimalt <LIZA.TZOTZOS@forces.gc.ca> **Cc:** Desormeaux LCdr C@C Navy D Nav P&T@Ottawa-Hull <Chantal.Desormeaux@forces.gc.ca>; Gagné CPO 2 YS@NPTG HQ@Esquimalt <Yannick.Gagne2@forces.gc.ca> **Subject:** FW: Stability - Final Answer Request Hi Ma'am, So, I looked at the JTAR and the various notes and gave everything some deep consideration here is the direction regarding the PO1 RQ: PO 002 Will Remain in the PO1 RQ PO 003 (Docking and Undocking) will Remain in the PO1 RQ in its entirety. PO 004 EO 004.01-Act as Electrical Manager for Emergency/Action Stations is to be moved to the Elect Supervisor USQ. PO 004 EO 004.02 - Act As DCPO is to be moved to the Mech Supervisor USQ. PO 004 EO 004.03 will be common to both of the PO2 USQs. This doesn't mean that it is to be moved to the PO2 RQ but needs to be present in both the Mech and Elect Supervisor USQs. Note The Role of the DCPO/ DCO on the MCDVs will be covered off by a separate piece of training (TBD if required).

Date	Decision/Event	Detail
		PO 005 Contains a couple of things which are more expansive than indicated in the QSP Managing Risk Assessment (AFT0170) – this needs to be broken out and taught as part of a larger Naval Materiel Assurance topic Managing short work periods (AFT0210) - This is one of the key tasks that PO1s and CPO2s will carry out. As well there are a few Tasks which are not covered at all: These are not learned properly through experience alone and need to be included at the PO1 level to prepare the member their role as CHOD. Note: Some of these topics are also part of the WEng training for the CPO2. I attach it as an example.

AET1010	Coordinate training and readiness – Both Personnel and Materiel (ie CRRs)/ captured at CPO2 Level needs to move to PO1 Common
AFT0080	Manage marine engineering trials agenda – Readiness / appears at PO1 Maint Manager, Needs to be added to PO1 Common
AFT0090	Manage maritime evaluation activities- appears at RSB Ops coord Needs to be added to PO1 Common
AFT0100	Manage material certification program – This is also in Naval Materiel Assurance/ Appears at PO1 Ship Sub Support Mgr Needs to be PO1 Common
AFT0820	Advise on particularized maintenance & repair schedule (PMRS) – Refits/ dockings/ NT in Elect Mgr. Needs to be PO1 Common
AFT0840	Advise on ship's standard maintenance & repair schedule (SSMRS)- Refits/ dockings/ is NT , should be T
AKT1200	Manage E&M inspections Readiness / is NT , should be T
AKT1240	Manage operational deficiency report requirement Readiness / is NT , should be T
AT0180	Coordinate equipment and system inspection Readiness/ is NT , should be T
AT0190	Coordinate systems trials Readiness / is NT , should be T
ZT0070	Manage notice for power Readiness / NT at Mech Mgr should be T PO1 Common

Date	Decision/Event	Detail
		Finally; The stability in the PO1 RQ serves the DCPO function as well as the CPO2 Dept Coord function. The level that exists in the PO2 RQ will feed into both the PO2 Mech Supervisor training and the PO1 RQ nicely. I do not feel that the calculations or concepts will be troublesome for the Electrical Sub Occ as they will be accustomed to much more complicated mathematics.

Lessons Learned:

1. IT assets. Access to more IT assets such as DWAN drops, keyboards and mouse is beneficial.
2. Experienced board members. Continuity of board members from related boards (i.e. RQ-PO2, JTAR) avoided duplication of course content and ensured the training was sequential in progression.
3. Assessment before TPts. Figuring out assessments first saved time and made it easier to identify critical teaching points.
4. TM availability. Having the training manager available for questions and clarifications on short notice was helpful to maintain the board's flow.
5. Stakeholder input. The stakeholder providing input when available was helpful to keep the board within scope and on mandate.
6. Annex F. Having the secretary capture acronyms and abbreviations throughout the board made it easier to log them than trying to capture them after the fact.
7. Standardized annex G timings table. Fill in resources and timings as the board creates EOs to have a good idea about time tables throughout and simplify completion of Ch. 1.
8. Review. Reviewing the document with all SMEs present helps to capture gaps. Though it seems tedious, it answers questions we didn't even know existed until we reread the document.

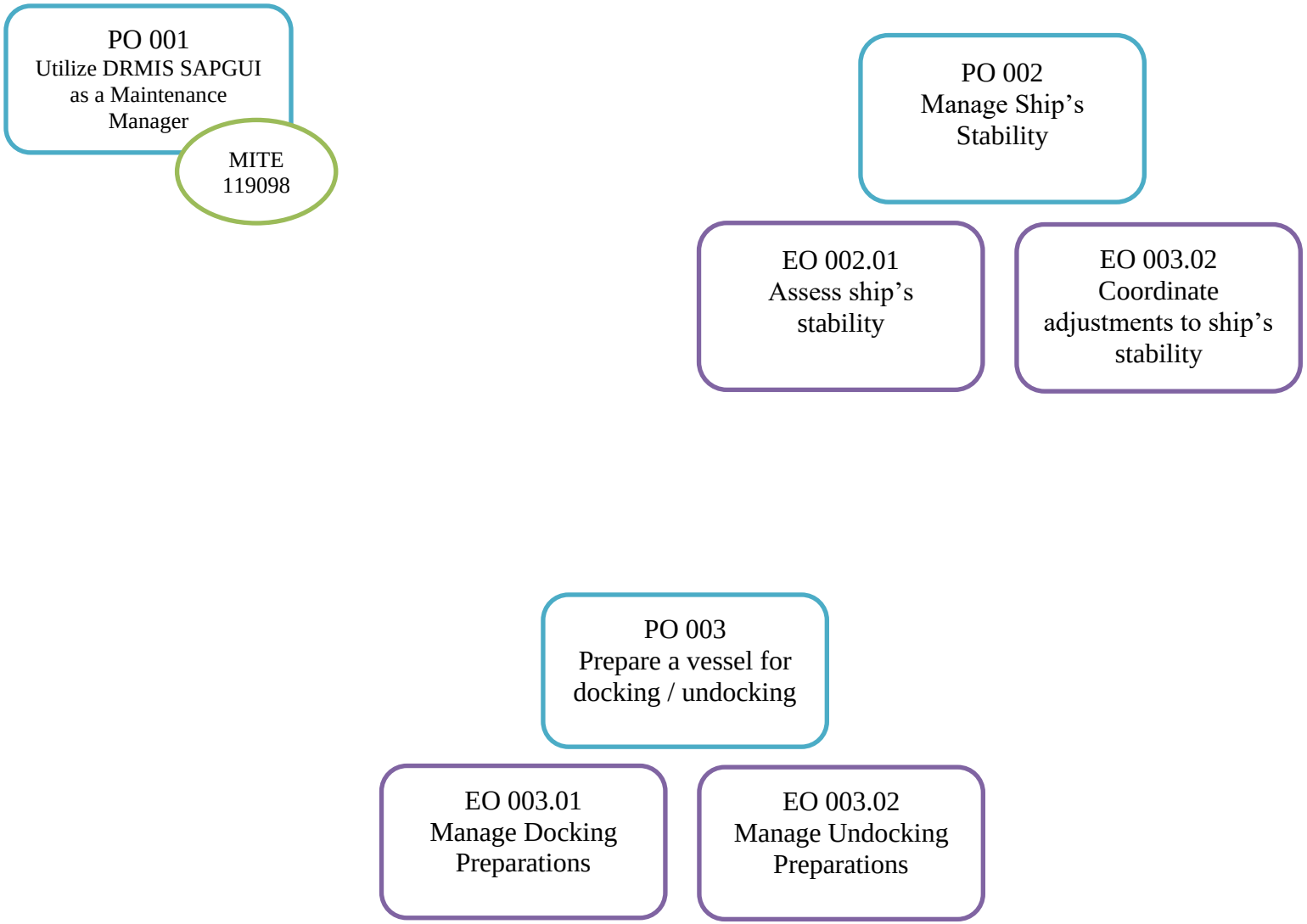
Guidance to Instructional Staff:

1. Guidance and relevant legacy documents for developer use is captured throughout the "Discussions/Decisions" above (highlighted in yellow).
2. Specific guidance for instructors is included in the "Remarks" paragraphs of each EO.

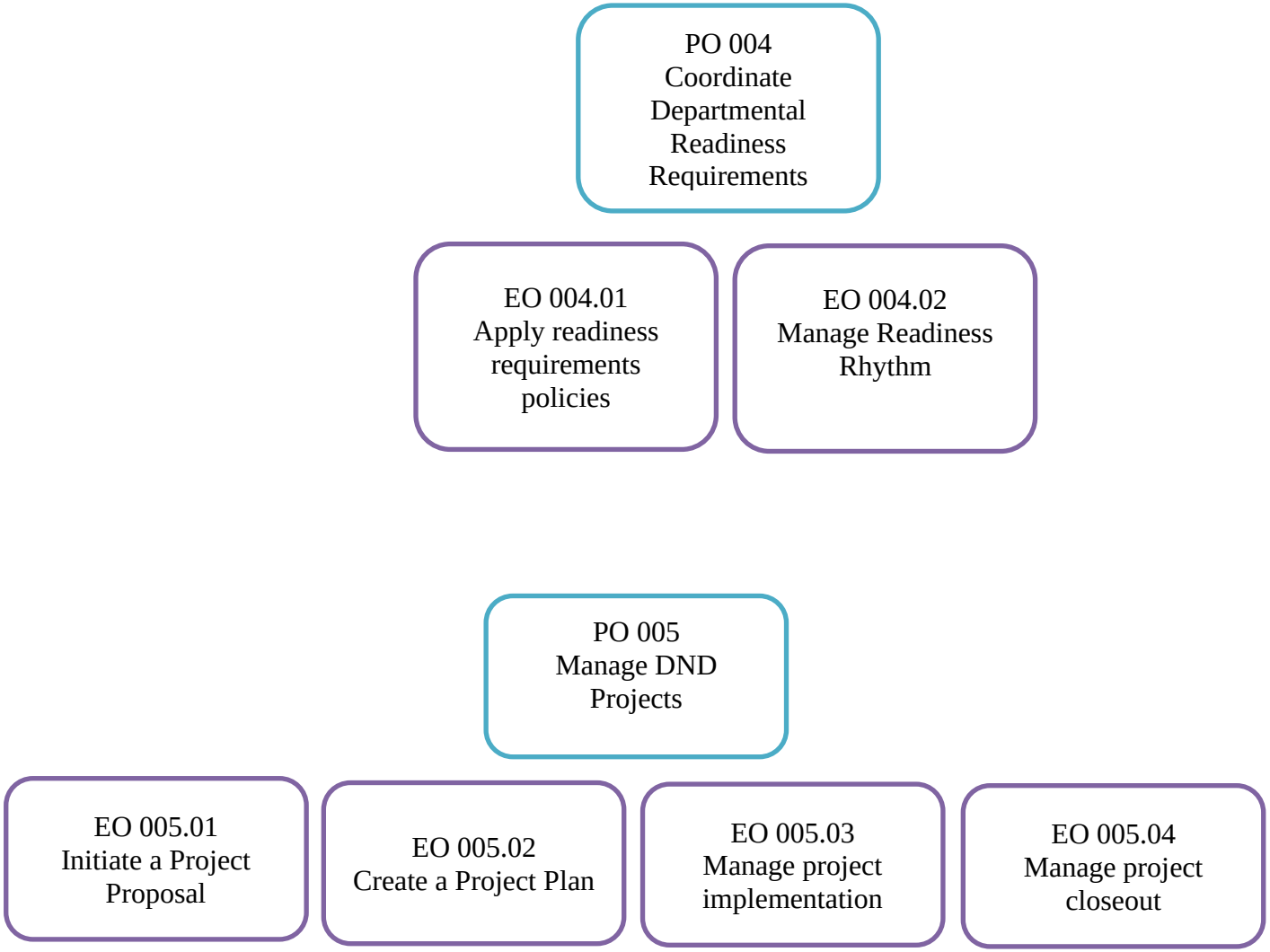
Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	RQ-PO2 QSP updated to include introduction to stability during SBT I/C PO to ensure Elec stream has the basics prior to RQ-PO1	Board Chair	22 Mar 18
2	Learning support requests (LSRs) completed for all TEL in Annex L	TDO	22 Mar 18
3	NCR tour approved	TM	30 Apr 18
4	Annex I “Costing Instructions” completed	NTDC(P) MS Div	31 May 18
5	Annex A “Main References” completed for development	NTDC(P) MS Div MAR TECH CCD Analysts	30 Jun 18
6	SPQR to update the task list with relevant tasks	TDO / TM to send to DPGR	30 Apr 18

ANNEX D - *QSPWB SCALAR*



MARPAC RDIMS #793521
Chap 1 approved DD MM YYYY
Chap 2, 3 approved DD MM YYYY


PO 004
Coordinate Departmental Readiness Requirements
EO 004.01
Apply readiness requirements policies
EO 004.02
Manage Readiness Rhythm
PO 005
Manage DND Projects
EO 005.01
Initiate a Project Proposal
EO 005.02
Create a Project Plan
EO 005.03
Manage project implementation
EO 005.04
Manage project closeout

ANNEX E - *ON JOB PERFORMANCE RECORD (OJPR)*

The MAR TECH RQ-PO1 course does not have an OJT component therefore this Annex has been left blank.

OJPR to be developed

ANNEX F - *ACRONYMS AND ABBREVIATIONS*

Acronym / Abbreviation	Term or Definition
AAR	After Action Report
BOI	Board of Inquiry
COA	Course of Action
CRR	Combat Readiness Requirements
DC	Damage Control
DNPS	Director Naval Platform Systems
DNR	Director Naval Requirements
DRMIS	Defence Resource Management Information System
DWP	Docking work period
EC	Engineering Change
F2F	Face to Face
FELEX	Halifax Class Modernization/Frigate Life Extension
FLOC	Functional Location
FTA	Fleet Technical Authority
GLM	General Loading Monitor
GT	Gas Turbine
HAT	Harbour acceptance trial
LCMM	Life-cycle Material Manager
MBS	Minimum Baseline Standard
MT&S	Manual of Trim and Stability
NavArch	Naval Architect
ND	Notification of Defect
NEM	Naval Environmental Manual
NFR	Naval Fleet Requirements
NETE	Naval Engineering Test Establishment
OPDEF	Operational Deficiency
OPI	Office of Primary Interest
PMO	Project Management Officer
PM	Project Manager
PMP	Project Management Professional
POC	Point of Contact
QAWC	Quality assurance work center
RF	Repair Facility
SAT	Sea Acceptance Trial
SME	Subject matter expert
SOR	Statement of requirement
SOW	Statement of work

STG	Sea Training Group
SWP	Short Work Period
TA	Technical Authority
TI	Technical Investigation
TOR	Terms of Reference
TRP	Tiered Readiness Program
WP	Work Period

ANNEX G - *SUGGESTED SEQUENCING AND TIMINGS*

1. This annex is to be used by Campus Support to properly schedule all required resources, including personnel, for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).
2. The table is completed in the preferred sequence for lessons and resources.
3. The embedded excel document includes calculations and can be used for any course amendments.

Annex G_Timings_v6_no stab

EO ###.##	EO Performance Statement	Personnel resources	Required Resources (for scheduling)	Trg Time per resource (mins)	EC Time (mins)	PC Time (mins)
~~**MITE 119098 PO 001: Utilise DRMIS SAPGUI as a Maintenance Manager**~~						
PO ATT:	~~**1195**~~	~~NOTE: Shall be scheduled at the beginning of the course. Can include loading of personnel that will not carry on to the MAR TECH Manager course.~~				
~~EO 001.01~~	~~Utilize DRMIS~~	~~DRMIS Instructor~~	~~Classroom w/ DWAN~~	~~1195~~	-	
PO 002: Assess Ship's Stability						
PO ATT:	**2065**					
EO 002.01	Assess ship's stability	Instructor	Classroom w/ DWAN	1200	115	
EO 002.02	Coordinate adjustments to ship's stability	Instructor	Classroom w/ DWAN	230		
EO 002.02	Coordinate adjustments to ship's stability	Instructor	BDCS/PCOT	200	120	200
PO 003: Prepare Vessel for Docking/Undocking						

PO ATT:	**1200**					
EO 003.01	Manage Docking Preparations	Transport	Transport	60		
EO 003.01	Manage Docking Preparations	Instructor/ Dock Master	DryDock tour	100		
EO 003.01	Manage Docking Preparations	Dock Master/ Instructor	Classroom w/ DWAN	100		
EO 003.01	Manage Docking Preparations	Instructor	Classroom w/ DWAN	750	60	
EO 003.02	Manage Undocking Preparations	Instructor	Classroom w/ DWAN	60		70
PO 004: Coordinate Departmental Readiness Requirements						
PO ATT:	**1165**					
EO 004.01	Apply readiness requirements policies	Guest lecturer - FTA	Classroom w/ DWAN	50		
EO 004.01	Apply readiness requirements policies	Teleconference with NFR Rep	Virtual Classroom or Classroom with VTC capabilities	50		
EO 004.01	Apply readiness requirements policies	Instructor	Classroom w/ DWAN	450	50	
EO 004.02	Manage readiness rhythm	Guest lecturer - N37	Classroom w/ DWAN	50		
EO 004.02	Manage readiness rhythm	Guest lecturer - SSO	Classroom w/ DWAN	100		
EO 004.02	Manage readiness rhythm	Guest lecturer - STG	Classroom w/ DWAN	75		
EO 004.02	Manage readiness rhythm	Instructor	Classroom w/ DWAN	75	50	215
PO 005: Manage DND Projects						
PO ATT:	**4755**	NOTE: Shall be scheduled after all other POs				

EO 005.01	Initiate a Project Proposal	Project Management SME	Classroom w/ DWAN	325	45	
EO 005.02	Create a Project Plan	Project Management SME	Classroom w/ DWAN	805	570	
EO 005.03	Manage Project Implementation	Project Management SME	Classroom w/ DWAN	250	220	
EO 005.04	Conduct project closeout phase	Instructor	Classroom w/ DWAN	100	440	
EO 005.04	Conduct project closeout phase	Instructor	NCR / NETE tour	2000		250
PO 006: Advise Command						
EO 006.01	Advise on Underway Equipment and System Failures	Instructor	Classroom w/ DWAN	100	900	
EO 006.02	Advise on Emergent Machinery/Equipment Failures	Instructor	Classroom w/ DWAN	200	100	
		Actual trg time:	**12130**		**Total mins:**	**14080**
PT:	**900**	**Course admin time:**	**1110**		**Total days:**	**35**

*If NCR tour is not approved, timings will change substantially as the ATT will be < 31 days, meaning less TAT and BAT allocated.

PROPOSED SCHEDULE FOR NCR VISIT

	SUN	MON	TUES	WED	THURS	FRI	SAT
AM (0800 to 1200)	Leave For Ottawa	Intro to NCR and DGMEPM	DNPS 7/8	DGNFD (DNR / DNCI / DNMCP)	D NAV P&T	NETE	Fly home from Montreal
Lunch (1200-1300)							

PM (1300-1600)	Arrive Ottawa	DNPS 2/3/6	DMEPM MSC/MWVA	DRMIS/ Group project time	WAR Museum Travel to Montreal	Industry (Gas Tops/L3 Mapps/ Schneider Electric)	Arrive Home Victoria

ANNEX H - *ADDITIONAL TRAINING RESOURCES*

All resources for MAR TECH Manager training are captured in Chapter 1 para 3-4 and Annex G. (**Annex H not to be deleted**)

ANNEX I - *COSTING INFORMATION*

Include information about cost in whichever format is applicable (**Not to be deleted**).

ANNEX J - *COURSE HISTORY*

This annex shall NOT be deleted. It must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details on course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications shall be captured in the table below. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Reason for amendment (with date)	Details on course history
QSPWB 5-16 Mar 18	The Marine Technician Manager RQ-PO1 is an amalgamation of elements of legacy Marine Engineer QL7, Electrical Technician QL6B, and Hull Technician QL6B. This course is a follow-on from the MAR TECH Supervisor RQ-PO2 course and MAR TECH PO2 USQ Mech and Elec. This course was developed with input from representatives from each of the three aforementioned legacy MOSIDs in an effort to ensure that the appropriate job elements were captured moving forward. The MAR TECH RQ-PO1 focuses on Project Management and HQ1 as this is the last RQ training for MAR TECH.
Post-QSPWB 19 Apr 18	After feedback from D Nav P&T it was determined that DCPO / HQ1 and Stability is a Mechanical responsibility and will not be covered at the RQ-PO1 level. POs 002 (Manage Ship's Stability) and 004 (Manage DCHQ) were removed making Project Management change from PO 005 to 003. Timings were updated accordingly. The following tasks were changed to NT at the RQ-PO1 level: AFT1520; IT0120; IT0130; IT0810.
	It was decided that undocking was not large enough to have its own EC therefore a homework assignment and questions on the PC would suffice to assess the trainees.
Post USQ PO2 Mech/Elec Supr Working Gp **7 – 11 May 18**	Further amendments are being made to the RQPO1 Mar Tech Mgr QSP after completion of the USQ PO2 Mar Tech Mech/Elec QSPWBs. Changes/amendments include: - Re-installment of PO 002 – Manage ship's stability (D Nav P&T Reqr) - Addition of PO 004 – Manage Personnel Requirements (decision from USQ PO2 Elec QSPWB). See Anx C for further details.
Post QSP WB 22 Oct 18	PO 006 Advise Command was created by the East Coast for addition to the QSP. QSP was amended under direction from D Nav P&T and MARTECH TM.
Post QSP 06 Dec 18	A meeting was held between NFS ((A) (P)), Coastal Advisor (P), MARTECH TM, and NTDC (Propel & P) to discuss the design of EO 002.01 – Assess Ship's Stability. It was decided that this EO would be based around Stability terminology and navigating the MT&S, vice the

	original design of having the trainee perform basic stability calculations. All meeting members unanimously agreed that only the Mechanical Stream members where the only personnel really required to perform the calculations. Electrical stream members where to be given basic guidance with regards to the calculations, but not perform them. Further discussion arose with regards to the similar stability terminology appearing on the RQ-PO2 course. It was decided that no changes would be made in regards to stability alignment at this time, as in its current state, it is better suited to aid legacy trades. Once the MARTECH specific training has fully absorbed the legacy training, the stability requirements and alignment within the MARTECH trade progression will be re-assessed.
14 Sep 22	Removal of DRMIS PO001 due to addition to RQPO2. DRMIS MOD 3 Maintenance Manager to be added as a Pre Requisite for RQPO1.

ANNEX K - *ASSESSMENT PLAN*

EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
EC 002.01	EO 002.01	Assess ship's stability	Scenario based assessment	75%	Remedial Training / Re-test
EO 002.02	EO 002.02	Coordinate adjustments to ship's stability	Using GLM to calculate stability and then brief MSEO on stability plan	Pass/Fail	Remedial Training / Re-test
PC 002	PO 002	Manage ship's stability	Scenario based assessment to develop a stability plan to brief MSEO	Stability is confirmed by BDCS to be +/- 3° port/stbd; draught +/- .25m LCF amidships based on trainee's recommendations.	First PC Fail results in a retest. Second PC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
EC 003.01	EO 003.01	Manage docking preparations	20 questions written exam; fill in applicable portions of CF 523; referencing required	80%	Remedial Training / Re-test

PC 003	PO 003	Prepare vessel for docking/undocking	30 questions written exam; fill in applicable portions of CF 523; no references	70%	Remedial Training / Re-test / TRB
EC 004.01	EO 004.01	Apply readiness requirement policies	Conduct a risk assessment on a scenario of an equipment failure and identify the subsequent OPDEF's priority	80% of checklist items	Remedial training/ re-test
EC 004.02	EO 004.02	Manage readiness rhythm	Develop the plan to execute one of a ship's CRR serials	70% of checklist items	Remedial training/ re-test
PC 004	PO 004	Coordinate Departmental Readiness Requirements	Draft a briefing note to the CO explaining the ship's current status in a 5 year cycle	70% of checklist items	Remedial Training / Re-test / TRB
EC 005.01	EO 005.01	Initiate a Project Proposal	Draft the response message (T -45) for the MSE section	100% on checklist	Remedial Training / Re-test
EC 005.02	EO 005.02	Create a Project Plan	Draft a Gantt chart and provide a project brief	70%	Remedial Training / Re-test
EC 005.03	EO 005.03	Manage project implementation	Modify timelines in Gantt chart from EC 003.02 given changing conditions	SAT all critical items	Remedial Training / Re-test
EC 005.04	EO 005.04	Manage project close-out	Draft a project AAR; provide project closeout brief	60% SAT	Remedial Training / Re-test

PC 005	PO 005	Manage DND Projects	Draft a memorandum detailing the progress of the identified Engineering Change. Provide an oral brief as a group to the assessor.	70% of checklist items	Remedial Training / Re-test / TRB

ANNEX L - *TEL PLAN*

EO	COA 1 TEL Delivery	COA 2 Interim Delivery	Impact if TEL not implemented - Realized benefits - E/W Coast Delta
002.01 / 002.02	**Stability Videos**: existing videos (YouTube) that demonstrate stability failures on various classes of civilian and military vessels. Updated videos can provide greater fidelity to stability factors thus increasing trainee understanding and application of concepts to calculations and recommendations. Proprietary considerations exist.	**VHS videos**: videos currently exist but are dated (B&W). Content remains unchanged but platforms and updates to today's technology. **NWO Stability Videos:** NWO uses videos to instruct stability. Developers will need to assess if these may be suitable to support instruction of these EOs	Trainees will not see modern ships affected by poor stability and damage. Poor quality of current videos impacts trainee retention as millennials expect and are used to far greater reality in animations and videos.
003.01/ 003.02	**Interactive stability digital modelling software**: software will demonstrate the effects to stability of various loading and docking/undocking changes to the ship (multiple platforms).	**Physical docking models**: see-through tank with ship and block models that can be filled and emptied to demonstrate docking. Ship's loads can be changed to demonstrate effects to stability. (currently available but not used given high level of instructor prep)	Trainees will not be able to properly visualize a docking or the effects of stability changes.
003.01/ 003.02	**CPF Docking Video**: CPF docking video with live footage and animations which shows what happens under the water line at various stages of docking/undocking.	**Stability photos**: existing photos that show the various stages of docking / undocking	Less realism; photos do not demonstrate the real-time docking / undocking process and do not include what happens under the water

ANNEX M - *TARGET POPULATION AND TRAINING ESTABLISHMENT ANALYSIS*

** The following information was compiled during the MAR TECH RQ-PO1 QSPWB Mar 2018 to give context for any civilian developers. It can be removed from the QSP if desired after initial course development.*

<u>Key Target Population Details</u> – MAR TECH RQ-PO1 candidates

"The learners' level of skill and knowledge with respect to the performance objectives (POs) is a factor which depends mainly on previous related training or education, and experience. It may also be affected by strong opinions or attitudes about the subject matter. Information should be obtained on the learners" -CFITES Vol 4

Characteristic	Target population details
General data	
• Potential group size • Expected annual throughput • Location(s) of the intended learners • Access to computers (job) • Gender • Age • Trades	• 6-12 (optimum 10) • 20-30 trainees (2-3 courses) per coast per year • East & West coast (very few fm inland); from ships / Repair facilities / Naval training system (NFS/NTDC) • 1 computer for every 2 PO2 onboard ship • ~80% male • 35-40 (varied); 15 yrs average in uniform; Merit loaded • MAR TECH Mechanical and Electrical
Learning skills and preferences	
Competencies (general)	Strong leadership; Critical thinking; Math; Communications (both); Administration; diplomacy / professionalism; Computer skills; Honour / Ethics; Technical skills
Reading and writing level	Intermediate (Tech Comm Mod C complete)
Preferences, attitudes towards specific type of learning material and equipment (i.e. TEL, aversion to lectures)	Interactive: doing versus sitting; case studies, scenarios; physical interaction with resources; social interaction with personnel (i.e. discussions); learn from each other
Educational level	High School grad with some post-secondary
Previous military training experience	MAR TECH RQ-PO2; pre-ILP, PLQ or equivalent
Current subject-matter competence	

Characteristic	Target population details
Experience with target performance	~~Exposure to HQ1 (managing more events compared to RQ-PO2);~~ Mech will have more exposure to stability; Very little if any to project management; 50% chance of having being part of docking/undocking
Ability to perform any part of the POs	Merit-loaded; shouldn't have issues
Perception of the impact of achieving the POs on self, work or career	Career-progression; required for promotion = high motivation
Job/career stage	Supervisor to Manager
Social Characteristics	
Relationship to peers	Familiarity with other candidates; peer-peer learning
Tendencies toward cooperation or competition	Team-oriented; rare competition

<u>Describing the learning environment</u> – NFS MS division

The "learning system" is composed of all factors that affect and are affected by the learning that takes place: learners (Target Population), instructional materials, the instructors, instructional equipment (such as computers, projectors, LMS, etc.), the instructional facilities, and the community or organization.

1. What are the characteristics of the teachers/trainers who will be using these materials?
What are the interests and preferences of the instructors? Sea-experience; should have instructor experience (min PO1); preferred to have a teaching aptitude *How do instructors feel about having instruction delivered via media or other non-traditional methods?* Just learning new techniques (not ppt); Should have AIT for increased familiarity *Do instructors like to use technology-based instruction?* Depends on age and familiarity with tech *What experience have instructors had with technology-based instruction?* BDCS/IPMS trainer; GLM; ppt; Office Suite; LabVolt; L-3 Degaussing trainer In the near future: welding trainer; stability trainer; DRMIS Project Builder
2. Are there existing courses/programs into which this piece of instruction must fit?
If so, what is the strategy used in these materials? DRMIS Maintenance Manager Similar to WENG RQ-CPO2 Project management Legacy Hull Tech QL6B; Mar Eng QL7; Electrician QL6B
3. What hardware is commonly available in the potential learning environments?
Are computer workstations available? Yes *How many?* 1 per candidate *DWAN, standalone?* DWAN *What software or other materials are available?* GLM; BDCS Trainer; MS Project (Licensing??); DRMIS (Project Builder)
4. What are the characteristics of the classes and facilities that will use the new instruction?
Set by Schedulers at the school